Subquadratic Sparse Attention: Content-Routed Exact Attention for Long Contexts

Jon Smirl

September 9, 2026

Abstract

Dense self-attention costs $O(n^2)$ in the sequence length n, which is the binding constraint on long-context language models. This paper presents *Subquadratic Sparse Attention* (SSA), an attention mechanism that, for each query, (i) routes to a small content-dependent set of key blocks using only per-block summary statistics, (ii) adds a local window, and (iii) performs *exact* softmax attention over the selected keys. The per-query work is $O(\kappa)$ in a fixed budget $\kappa \ll n$ plus a sublinear routing cost: a flat router runs the layer in $O(n\sqrt{n})$ (at block size $b = \sqrt{n}$, where the budget grows as $\sqrt{n}$), and a hierarchical router runs it near-linearly at fixed b and fixed κ—the configuration in which retrieval is also flat in n (Section 4.4 keeps the two regimes apart). A supporting theory establishes when this is sound. A retrieval-margin analysis shows that softmax attention recovers a target key with weight $\sigma(\beta\Delta - \log \mu)$, where Δ is the score margin and μ the number of competing distractors; selection works because it cuts μ from n to κ, making recovery *flat in n*. The analysis gives the admissible routing bound that licenses summary-only selection, a tempered (cumulant) routing score that—unlike centroid routing—sees in-block outliers, and a closed-form variance prune test from Samuelson's inequality. A limit is then proved: cheap and lossless selection cannot hold for arbitrary keys (a one-line probe argument); adding length-robustness gives the trilemma (at most two of the three). So SSA's subquadratic *exactness* is licensed by the *benign geometry* of trained representations—geometry that training can be made to manufacture, via a routability regularizer that shrinks off-target spread. Finally, length generalization (rotary position plus staged continued training reaches 32× the trained length at 0.98 recall for ∼800 adaptation steps) and a construction pipeline that converts a dense pretrained model into a subquadratic one by swapping the attention and briefly adapting (recovering to within +1.2 perplexity of a dense model given equal training while attending 38% of keys) are demonstrated. At operational scale, a fixed-beam center-radius router and sparse GQA execute every layer and head of a frozen Qwen2.5-0.5B over 10,000,128 tokens on one RTX Pro 6000 in 713.2 s with 25.72 GB peak allocation. A 4K dense-equivalence gate, dense and streamed 128K retrieval gates, and one semantic 10M retrieval ranking pass. The 10M result establishes the complete execution conjunction; one retrieval instance does not establish broad quality preservation. A separate adaptive reference bounds omitted softmax mass, the subset-to-dense KL divergence, and attention-output error from key and value summaries, with a full-scan fallback when the requested tolerance cannot be certified sparsely.

Evidence status

Claim	Status	Scope
Complete transformer execution beyond 10M	demonstrated	one frozen 0.5B model, all layers and heads, one GPU
Subquadratic router and sparse kernel at 10M	demonstrated	bounded fixed-beam tree; exact softmax over the selected set
Semantic retrieval at 10M	narrow evidence	one NIAH ranking; no broad task suite or dense 10M baseline
Near-floor kernel scaling to 12M	demonstrated	single-head synthetic IVF benchmark
Omitted-mass and output-error certificates	reference implementation	CPU, adaptive, worst-case full scan
Supporting routing invariants	proved in this paper	self-contained statements and proofs in Appendix B; private Lean audit is corroborating provenance
Cheap worst-case losslessness	impossible under the stated models	grounded probes have a b/n ceiling; a K-state index with unread-output width a has a $K(b+a)/n$ ceiling

The paper reports the system as it exists at the stated measurement points. Experiment fixtures are kept separate when they establish different claims; results from synthetic kernel timing, real-model routing quality, and complete-transformer execution are not combined into an unmeasured speedup or quality claim.

1 Introduction

A transformer layer computes, for queries $Q \in \mathbb{R}^{n\times d}$, keys $K \in \mathbb{R}^{n\times d}$, and values $V \in \mathbb{R}^{n\times d}$,

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\Big(\tfrac{QK^\top}{\sqrt{d}}\Big) V.$$

The $n \times n$ score matrix makes both time and memory $\Theta(n^2 d)$. For contexts of 10^6–10^7 tokens this is prohibitive, yet most of the matrix is near-zero: for a given query only a small set of keys carries appreciable weight. The question is whether one can *find* that set without forming all n^2 scores.

Three subquadratic families answer differently. *Kernel / linear attention* replaces $\exp(\langle q, k\rangle)$ by a factorizable feature map $\phi(q)^\top \phi(k)$, giving $O(n)$ cost but a low-rank (smoothed) attention matrix. *Two-pass structural factorization* mixes within blocks and then across equal within-block slots, sharing intermediates while retaining a path between every token pair. *Sparse / selective attention* keeps the exact softmax but evaluates it only on a chosen subset of keys. SSA is in the third family, with three design commitments:

1. **Content-dependent selection.** The chosen keys depend on the query, not only on position; this is what lets a query reach the one relevant block a million tokens back.
2. **Summary-only routing.** Selection uses per-block statistics (a mean, a covariance), so it does not read every key—that is where the subquadratic saving comes from.
3. **Exact attention over the selected set.** Within the selected keys the softmax is computed exactly, so there is no kernel-approximation error on the keys that matter.

The paper is organized around one tension. Selection is cheap only if it can judge a block from its summary; it is *correct* only if those summaries do not hide a key that mattered. Sections 3–5 make both sides precise; Section 6 shows they cannot both hold in the worst case, so something must give; Section 7 shows what gives—the data geometry, which training shapes. Sections 8–9 turn the mechanism into a usable recipe.

2 Notation and the retrieval view

Fix a single query $q \in \mathbb{R}^d$ and keys $k_1, \ldots, k_n$. Write the *logit* (score) of key j as

$$a_j \;=\; \langle q, k_j \rangle, \qquad w_j \;=\; \frac{e^{\beta a_j}}{\sum_{l=1}^{n} e^{\beta a_l}}, \qquad \beta = \frac{1}{\sqrt{d}},$$

so w is the attention distribution and β its inverse temperature. The output is $o = \sum_j w_j v_j$.

A long-context layer is, operationally, an *associative recall*: a query must place most of its weight on the key(s) that hold the information it needs and little on the rest. Attention is therefore analyzed by the weight it puts on a designated *target* key $k_\star$ relative to the *distractors* $\{k_j\}_{j \neq \star}$.

3 Retrieval margin and the recovery weight

Define the *margin* of the target as the gap between its logit and the typical distractor logit,

$$\Delta \;=\; a_\star - \bar{a}_{\text{dist}}, \qquad \bar{a}_{\text{dist}} = (\text{typical } a_j,\ j \neq \star).$$

Suppose there are μ effective distractors with logit $\approx a_\star - \Delta$. Then the target weight is

$$w_\star \;=\; \frac{e^{\beta a_\star}}{e^{\beta a_\star} + \mu\, e^{\beta(a_\star - \Delta)}} \;=\; \frac{1}{1 + \mu\, e^{-\beta\Delta}} \;=\; \sigma(\beta\Delta - \log \mu), \tag{3.1}$$

where $\sigma(x) = 1/(1+e^{-x})$ is the logistic. Equation (3.1) is the *recovery weight*. Two consequences:

- **Detectability threshold.** The target is recovered ($w_\star > \frac{1}{2}$) iff

$$\boxed{\beta\,\Delta \;>\; \log\mu} \tag{3.2}$$

  Recovery degrades only *logarithmically* in the number of distractors.

- **Why selection helps, and why it is flat in n.** Dense attention pays $\mu = n$. A selector that attends only a budget of κ keys pays $\mu = \kappa$, replacing the threshold $\log n$ by $\log\kappa$. If κ is held *fixed* as the context grows, the threshold (3.2) does not move with n: retrieval accuracy becomes *independent of context length*. This is the entire value proposition of selection, and it is why a fixed-budget selector can answer single-target queries at 10^6–10^7 tokens that dense attention, with its $\log n$ erosion, increasingly struggles with.

The catch is hidden in the phrase "a selector that attends κ keys": the selector must *contain the target in its budget.* Sections 4–6 are about exactly that.

4 The algorithm

4.1 Block partition and summaries

Partition the n keys into B contiguous *blocks* of size b (so $Bb = n$). For block c precompute, once, the summary statistics

$$\mu_c = \frac{1}{b}\sum_{j\in c} k_j \quad \text{(mean)}, \qquad \Sigma_c = \frac{1}{b}\sum_{j\in c}(k_j - \mu_c)(k_j - \mu_c)^\top \quad \text{(covariance)}, \tag{4.1}$$

and a radius $R_c = \max_{j\in c}\|k_j - \mu_c\|$. In practice Σ_c is kept diagonal, $\sigma_c^2 = \operatorname{diag}\Sigma_c$, so a block's summary is $O(d)$ numbers. (The diagonal form is fine for the *heuristic* routing score, but it voids the Samuelson prune *certificate* unless the surrogate bound is used—see the diagonal-summary caveat in Section 5.3.)

4.2 Routing score

For a query q, the *routing score* of block c estimates the largest logit the block can contribute. The exact quantity is the block's log-sum-exp $\mathrm{LSE}_c(q) = \log\sum_{j\in c} e^{\beta\langle q,k_j\rangle}$, which requires reading every key. SSA uses its *second-order (cumulant) surrogate*, computable from the summary alone:

$$r_c(q) \;=\; \langle q,\mu_c\rangle \;+\; \tfrac{\beta}{2}\,q^\top\Sigma_c\,q \quad\approx\quad \langle q,\mu_c\rangle + \tfrac{\beta}{2}\,\langle q^2,\sigma_c^2\rangle. \tag{4.2}$$

Equation (4.2) is the Taylor/cumulant expansion of the tempered mean $\beta^{-1}\log\frac{1}{b}\sum_{j\in c} e^{\beta\langle q,k_j\rangle}$ about $\beta = 0$: the first cumulant is the mean logit $\langle q,\mu_c\rangle$; the second adds the *variance of the* *logit across the block*, $q^\top \Sigma_c q$. The variance term is what makes routing see an in-block outlier (Section 5.2).

4.3 Selection and exact attention

For each query q_i (at position i):

1. Compute $r_c(q_i)$ for all blocks c whose keys are causally visible (c ends at or before i).
2. Select the top-k blocks by r_c, *union* a local window of the w most recent blocks (recency is always relevant and cheap to include), giving a key set S_i with $|S_i| = \kappa \le (k+w)b$.
3. Compute exact softmax attention restricted to S_i:

$$o_i \;=\; \sum_{j\in S_i} \frac{e^{\beta\langle q_i,k_j\rangle}}{\sum_{l\in S_i} e^{\beta\langle q_i,k_l\rangle}}\, v_j. \tag{4.3}$$

```
Algorithm 1: SSA forward (one head)
Input:  Q, K, V in R^{n x d};  block size b;  budgets k (global), w (local)
Precompute: for each block c, mean mu_c and diagonal covariance Sigma_c     // O(n d)
For each query q_i  (in parallel over i):
    Routing:    for each causally visible block c:
                    r_c <- <q_i, mu_c> + (beta/2) * <q_i^2, diag Sigma_c>
    Selection:  T   <- (top-k blocks by r_c)  union  (w most-recent blocks)
    Causal:     S_i <- keys of blocks in T with position <= i
    Attention:  o_i <- softmax_{j in S_i} ( <q_i,k_j> / sqrt(d) ) * V[S_i]   // exact
Return O
```

The final line's original-position cut is logically necessary, not cosmetic. A selected set reused wholesale at every query is non-causal whenever it contains a position later than the query. Intersecting it with $\{j : j \le i\}$ makes the relation causal for every proposed set, and compositions of such causal relations remain causal. The implementation additionally applies a token-level causal mask, so even a selected block that straddles the query cannot expose its future tokens; chunk-level causality alone would not suffice.

4.4 Complexity—two regimes, and which claim lives where

Routing is $O(n\,B\,d)$ if every query scores every block, and attention is $O(n\,\kappa\,d)$. Two parameter regimes must be kept apart, because the cost claim and the retrieval-flatness claim of Section 3 do not live in the same one:

- **Cost-optimal flat router**, $B = \sqrt{n}$, $b = \sqrt{n}$, fixed block-counts k, w: routing $O(n^{1.5}d)$, attention $O(n\,\kappa\,d) = O(n^{1.5}d)$; total $O(n^{1.5}d)$—already a large saving over n^2. But here the key budget $\kappa = (k+w)\sqrt{n}$ *grows with* n: the detectability threshold $\log\kappa$ of Section 3 rises as $\frac{1}{2}\log n$ (the $\log n$ erosion returns at half rate), and the $1/b$ mean attenuation of Section 5.2 worsens as $1/\sqrt{n}$. Subquadratic, but *not* retrieval-flat.

- **Retrieval-flat flat router**, b fixed (the measured setting, $b = 128$), fixed k, w: κ is fixed, so recovery is flat in n (Section 3)—but scan-all-blocks routing is $O(n\,B\,d) = O(n^2 d/b)$, quadratic up to a constant (the measured $n^{1.76}$ router and $n^{2.12}$ maskbuild walls of Section 10).

- **Hierarchical router at fixed b and fixed κ**—the configuration that delivers both. Organize blocks into a tree of summaries and descend it per query with branch-and-bound pruning (Section 5.1), so each query inspects $O(k\log B)$ nodes rather than all B—a benign-geometry cost, not a worst-case guarantee. Routing drops toward $O(n\log n\,d)$ while κ stays fixed. Two practical approximate instances are measured: the IVF router of Section 10 (0.93–0.97 block agreement), which places the isolated 12M kernel at $\sim$2.9× the $n\,\kappa$ floor, and the fixed-beam center-radius tree used in the complete 10M transformer. Both occupy the cheap+length-robust-but-approximate corner of Section 6, not the certified one.

So "$O(n\sqrt{n})$ per layer" and "recovery flat in n" are claims about *different* configurations, and the hierarchical (or IVF) router at fixed b, κ is what reconciles them. Either way the dominant n^2 term is gone. In a block-sparse kernel implementation the practical speedup over a dense exact kernel was 20.6× at $n = 262{,}144$ on a single accelerator (Section 11).

Structural-factorization comparison. Write a position as $(\text{block}, \text{slot})$ with block width w and n/w blocks. A within-block pass can move from a source to the target's slot inside the source block; a same-slot cross-block pass then reaches the target. Thus two non-dense passes give complete two-hop connectivity. If—and only if—a cost model supplies work proportional to $n(w + n/w)$, AM–GM gives $w + n/w \geq 2\sqrt{n}$, uniquely at $w = \sqrt{n}$. This is not SSA's omission-based mechanism, and complete reach is not dense-attention equivalence. With the within-block pass fixed, varying the cross-block matrices has parameter-space dimension at most $w(n/w)^2$, strictly below the n^2 target dimension when $w > 1$; the symmetric statement holds with the other pass fixed. This proves neither a limitation with both passes free nor a rank bound—the product can have full rank.

There is a similarly clean but deliberately unimplemented fanout calculation. If expanding a tree node of fanout f costs exactly f scalar units per level, the path cost is proportional to $f\log_f B = (\log B)f/\log f$, minimized continuously at e and among integers $f \geq 2$ at 3. GPU vectorization, memory traffic, fixed-beam recall, and build cost are absent from that model, so it is a benchmark hypothesis rather than a reason to replace the measured 10M configuration's fanout 16.

5 Routing theory: when a summary is enough

The routing score (4.2) is a *heuristic* surrogate. This section gives the *certified* objects—upper bounds that make selection provably lossless—and the prune test SSA uses.[1]

5.1 The admissible bound and branch-and-bound exactness

For any key k in block c, decompose its logit around the block mean and apply Cauchy–Schwarz:

$$\langle q, k\rangle = \langle q, \mu_c\rangle + \langle q, k-\mu_c\rangle \;\le\; \langle q, \mu_c\rangle + \|q\|\,\|k-\mu_c\| \;\le\; \underbrace{\langle q, \mu_c\rangle + \|q\|\,R_c}_{U_c(q)}. \tag{5.1}$$

$U_c(q)$ is an *admissible upper bound*: no key in block c has a logit exceeding it, and it depends only on the summary (μ_c, R_c). This licenses exact selection at adaptive cost.

> **Branch-and-bound selection.** Maintain the best *actual* logit found so far, $s^\star$. Open blocks in decreasing $U_c(q)$. Stop as soon as the next block has $U_c(q) \le s^\star$: by (5.1) no unopened block can contain a key beating $s^\star$. The result is identical to scanning all keys, but only the blocks whose bound clears $s^\star$ are ever read.

The cost of branch-and-bound is the number of blocks whose bound exceeds the true best—a quantity governed entirely by how *tight* the bound (5.1) is, i.e. by the geometry of the keys (Sections 6–7).

The hierarchy also has a monotone pruning law. Against a fixed reference point, a descendant ball's reach is no larger than its ancestor's, hence its score cap is no larger. At any fixed threshold, every region dropped by the ancestor cap is therefore also dropped by the descendant cap: refinement can only enlarge the certified drop set. This is a correctness/order statement, not a complexity theorem—an algorithm may still open every node, and region counts need not equal key counts or wall time.

Anisotropic refinement. The isotropic radius R_c is loose when a block's keys are spread unevenly across directions. Using the covariance ellipsoid instead,

$$\langle q, k-\mu_c\rangle \;\le\; \sqrt{q^\top \Sigma_c\, q}\cdot \rho_c, \qquad \rho_c = \max_{j\in c} \left\|\Sigma_c^{-1/2}(k_j-\mu_c)\right\|, \tag{5.2}$$

[1] These supporting results—the admissible and ellipsoidal search bounds, the Samuelson prune gate, the tempered log-partition sandwich, the recursive-radius correctness of hierarchical pruning, the value-aware drop bound, and the no-free-selection limit of Section 6—are machine-checked in a Lean 4 formalization (axiom-pure, no `sorry`) [17], which is maintained separately and is not part of this repository's artifact. The formalization is optional audit provenance: the public mathematical arguments are given in this paper, with the complete 10M routing invariants reproduced in Appendix B.

which replaces $\|q\|R_c$ by the *directional radius* $\sqrt{q^\top \Sigma_c q} \cdot \rho_c$. When the keys are anisotropic (the usual trained case), $\sqrt{q^\top \Sigma_c q}$ can be far smaller than $\|q\|R_c$ for queries pointing along thin directions—a tighter, query-specific bound, and exactly the quantity the cumulant score (4.2) already trades on.

The regularised form is not the same bound. Equation (5.2) is admissible in exact arithmetic, but Σ_c is singular whenever a block holds fewer keys than the head dimension—the usual case—so every implementation forms $S_c = \Sigma_c + \varepsilon I$ and takes $\rho_c = \max_j \|S_c^{-1/2}(k_j - \mu_c)\|$. *The quadratic form must then be S_c's as well*: Cauchy–Schwarz reads $\langle q, k - \mu_c\rangle = \langle S_c^{1/2} q, S_c^{-1/2}(k - \mu_c)\rangle \le \sqrt{q^\top S_c q} \cdot \rho_c$, and $q^\top S_c q = q^\top \Sigma_c q + \varepsilon\|q\|^2$. Pairing a radius measured in the S_c metric with the *unregularised* $\sqrt{q^\top \Sigma_c q}$ yields a quantity that can fall below the block's own maximum, and a bound that does so can discard the block holding the argmax—at which point losslessness is no longer a property of the construction. The deficit is small and direction-dependent: on 32-dimensional blocks of ≈ 33 keys it appeared on 5 of 128 block–query pairs, by at most 2.8×10^{-3}, and *only* for queries aligned with a block's leading direction. That last clause is the practical warning, because it is the realistic case—a router's queries are not orthogonal to the keys they route to—and it is why a random-query admissibility test at a single geometry does not detect the error.

5.2 Why second order: centroid routing is blind to outliers

Centroid routing uses only $r_c = \langle q, \mu_c\rangle$. A single key $k_\star$ in a block of size b contributes $\frac{1}{b}k_\star$ to μ_c, so its signal in the mean is attenuated by $1/b$: a lone target in a large block is invisible to centroid routing. The variance term in (4.2) repairs this, because an outlier aligned with q inflates $q^\top \Sigma_c q$. Concretely, if block c holds one key with $\langle q, k_\star\rangle = a_\star$ and $b-1$ keys with logit ≈ 0, then $\langle q, \mu_c\rangle \approx a_\star/b$ but $q^\top \Sigma_c q \approx a_\star^2/b$, so the second-order score $r_c \approx a_\star/b + (\beta/2)a_\star^2/b$ carries the quadratic outlier signal the mean discards.

The tempered family and its bias. It is the *normalized* tempered mean $g_c^{(\beta)} = \beta^{-1} \log \frac{1}{b} \sum_{j\in c} e^{\beta\langle q, k_j\rangle}$—the object expanded in (4.2)/Appendix A.2—that interpolates between the mean logit ($\beta \to 0$) and the block max ($\beta \to \infty$). The unnormalized log-sum-exp $r_c^{(\beta)} = \beta^{-1} \log \sum_{j\in c} e^{\beta\langle q, k_j\rangle} = g_c^{(\beta)} + (\log b)/\beta$ differs from it only by a block-size constant (irrelevant when ranking equal-size blocks, where the two orderings coincide) and is sandwiched by

$$\max_{j\in c}\langle q, k_j\rangle \;\le\; r_c^{(\beta)} \;\le\; \max_{j\in c}\langle q, k_j\rangle + \frac{\log b}{\beta}. \tag{5.3}$$

So the tempered score estimates the block's *best* logit—exactly what selection wants—with bias at most $(\log b)/\beta$. Small β smooths over outliers (the centroid failure); large β removes the bias but amplifies noise and loses the averaging that makes summaries stable. The cumulant form (4.2) is the second-order Taylor truncation of $g_c^{(\beta)}$ about $\beta = 0$, used at the measured optimum $\beta \approx 2$. This approximation now has a deterministic interval certificate. If every in-block logit $x_j = \langle q, k_j \rangle$ lies in $[\ell, h]$ and $K(\beta) = \log \frac{1}{b} \sum_j e^{\beta x_j}$, then

$$\left| K(\beta) - \left(\beta \bar{x} + \frac{\beta^2}{2} \operatorname{Var}(x) \right) \right| \leq \frac{|\beta|^3 (h - \ell)^3}{6}. \tag{5.4}$$

For $\beta \neq 0$, divide by $|\beta|$ to bound the error of $g_c^{(\beta)}$. The proof differentiates the finite log-partition three times and applies Taylor's theorem: its third derivative is the tilted third central moment, in magnitude. The range bound works because every tilted mean remains in $[\ell, h]$, so the absolute centered cube is at most $(h - \ell)^3$ throughout the entire segment from 0 to β. Controlling the ordinary third cumulant only at zero does *not* certify the remainder. Thus an Edgeworth term measured at zero remains a heuristic, while (5.4) is a valid, often loose, blockwise interval. The residual is one mechanism behind the isolated-needle failures measured in Section 11.

5.3 The Samuelson prune test

A closed-form, summary-only test for *discarding* a block uses *Samuelson's inequality*: for any reals $s_1, \ldots, s_m$ with mean $\bar{s}$ and population variance $\operatorname{Var} = \frac{1}{m} \sum_j (s_j - \bar{s})^2$, every element obeys

$$(s_i - \bar{s})^2 \;\leq\; (m-1) \operatorname{Var}, \qquad \text{equivalently} \qquad \max_j s_j \;\leq\; \bar{s} + \sqrt{(m-1) \operatorname{Var}}. \tag{5.5}$$

Apply it to the in-block logits $s_j = \langle q, k_j \rangle$, whose mean is $\langle q, \mu_c \rangle$ and whose variance is $q^\top \Sigma_c q$. Then the block's best logit is bounded by $\langle q, \mu_c \rangle + \sqrt{(b-1)\, q^\top \Sigma_c q}$, giving the *prune gate*: block c can be safely discarded against a threshold $\tau = s^\star$ whenever

$$\boxed{(s^\star - \langle q, \mu_c \rangle)^2 \;>\; (b-1)\, q^\top \Sigma_c q \quad \text{and} \quad \langle q, \mu_c \rangle < s^\star} \tag{5.6}$$

i.e. when the *margin* of the current best over the block mean exceeds $\sqrt{(b-1) \cdot \text{spread}}$. The test needs only (μ_c, Σ_c). It is *sufficient* (it never wrongly prunes) but not necessary—though by Proposition 1 it is the *tightest* test computable from those statistics, so its non-necessity is a property of summary-only routing and not a slack in this particular gate. It sharpens the radius bound (5.1) when the block's spread along q is small relative to its worst-case radius ($\sqrt{(b-1)\, q^\top \Sigma_c q} \ll \|q\| R_c$); neither bound dominates in general—$\sqrt{(b-1)\, q^\top \Sigma_c q}$ can exceed $\|q\| R_c$ by up to a factor $\sqrt{b-1}$ for spread-out blocks—so the implementation takes the minimum of the two. Equation (5.6) is the operational core of cheap exact selection: it fires—and the block is skipped—precisely when the off-target spread $q^\top \Sigma_c q$ is small, which is the benign-geometry condition of Section 7.

Diagonal-summary caveat. (5.5)–(5.6) are sound with the *full* quadratic form $q^\top \Sigma_c q = \mathrm{Var}_{j\in c}\langle q, k_j\rangle$. With the diagonal summary of (4.1), the proxy $\langle q^2, \sigma_c^2\rangle$ *under-estimates* the true logit variance whenever cross-covariances are positive, and the gate can then wrongly prune: **run on diagonal summaries,** (5.6) **is a heuristic, not a certificate.** A sound $O(d)$-summary surrogate exists: by the triangle inequality in L^2 over the block, $\mathrm{Var}_{j\in c}\langle q, k_j\rangle \le \big(\sum_i |q_i|\, \sigma_{c,i}\big)^2$, so substituting $(\sum_i |q_i|\, \sigma_{c,i})^2$ for $q^\top \Sigma_c q$ in (5.6) keeps the gate admissible at the same summary cost (looser, so it fires less often). The prune-rate measurements reported in this paper compute the per-member logit variance directly—the full quadratic form—so they certify the full-covariance gate, not the diagonal shortcut.

5.4 The summary-only floor: Samuelson is the tightest, and how far it is from the partition's own limit

Equation (5.6) is labelled *sufficient.* That labelling is right, and it leaves an obvious question unanswered: could a *cleverer* function of the same summary prune more? It could not, and the reason is that Samuelson's inequality is not merely valid but *attained.*

Proposition 1 (Samuelson is the tightest summary-only bound)**.** *Fix $m \ge 2$, a mean $\bar{s}$ and a population variance* $\mathrm{Var} > 0$*. The configuration*

$$s_1 = \bar{s} + \sqrt{(m-1)\,\mathrm{Var}}, \qquad s_2 = \cdots = s_m = \bar{s} - \sqrt{\mathrm{Var}\,/(m-1)} \tag{5.7}$$

has exactly mean $\bar{s}$ and variance Var*, and its maximum equals $\bar{s} + \sqrt{(m-1)\,\mathrm{Var}}$. Consequently any bound $U < \bar{s} + \sqrt{(m-1)\,\mathrm{Var}}$ is violated by a block whose summary (μ_c, Σ_c, b) the router cannot distinguish from the one it read. No admissible bound computable from (μ_c, Σ_c, b) alone is tighter than* (5.5)*.*

The arithmetic is immediate from (5.7) and is verified numerically at $m \in \{2, 4, 8, 16, 64, 256\}$ in `ssa/bound_floor.py`. The content is the quantifier: the bound is not conservative *given the summary*, so every improvement in pruning must come from a better *partition* or from *reading keys*—never from a better formula on the same statistics. This is the necessary-side companion to the benign-geometry condition (7.1), which is sufficient only.

What the partition costs, and what the summary costs. Proposition 1 makes a decomposition measurable. For a fixed query and partition, the tightest admissible bound that exists at all is the *oracle* $U_c = \max_{k\in c}\langle q, k\rangle$; it is not a router (computing it reads the block), but it is the floor the partition imposes on *any* correct branch-and-bound. Then

$$\underbrace{\mathrm{cost(oracle)}}_{\text{price of the partition}} \le \underbrace{\mathrm{cost(ellipsoidal)}}_{\text{reads } \rho_c} \le \underbrace{\mathrm{cost(Samuelson)}}_{\text{summary-only}}.$$

Measured on clustered synthetic keys ($n = 4096$, $d = 64$, $B = 64$ k-means blocks, 120 queries per row, all four bounds admissible so recall is 1.000 throughout):

spread	oracle	ellipsoidal	Samuelson	summary ÷ floor
0.02	89.0	203.2	720.3	8.1×
0.05	81.0	543.6	975.4	12.0×
0.10	74.6	1765.9	2165.0	29.0×
0.20	70.2	3564.3	3685.8	52.5×
0.40	64.1	4022.8	4045.3	63.1×

Three readings, in decreasing order of how much they change the picture. First, the routability program (7.2) has a *measurable ceiling*: driving the geometry benign moves the summary price from 63× the floor to 8×, monotonically—but not to 1×, and Proposition 1 says why it cannot. Second, *reading keys is worth roughly* 3.5× *at benign geometry* (203.2 against 720.3 at spread 0.02), which is a quantitative justification for the anisotropic refinement (5.2); ρ_c is not a minor sharpening but most of the distance to the floor—*at these shapes*, a qualification the next paragraph makes precise. Third, the partition price is nearly flat in the spread ($89.0 \to 64.1$) while the summary price moves by a factor of six: *the geometry is a fact about summaries, not about partitions.*

ρ_c inverts when a block holds far fewer keys than dimensions. The table is $d = 64$ with blocks of ≈ 64 keys. On real Gemma-2 layer-6 keys ($d = 2304$, blocks of 64–256) the same stored scalar *costs* rather than buys—0.7×, 0.9×, 1.0× at $n = 4096, 16384, 65536$—because Σ_c is then deeply rank-deficient, $S_c = \Sigma_c + \varepsilon I$ is dominated by ε, and the honest $\sqrt{q^\top \Sigma_c q + \varepsilon \|q\|^2}$ is loose exactly where the whitened radius is large. The anisotropic refinement is therefore a recommendation *scoped to $b \gtrsim d$*. Omitting the regularization term produces an inadmissible estimate and the wrong ordering of the two bounds on real keys.

Scope. These are k-means blocks on synthetic clustered keys at one (n, d, B) and one query-noise level, which is the adaptive/IVF regime, not the contiguous-position blocking of the flat kernel; the oracle is a reference and not an achievable router; and the floor is a statement about *lossless* selection, so a budget-κ lossy router may sit below it and SSA's does. Proposition 1 itself is geometry-free and carries none of these caveats.

Ellipsoidal refinement and the remaining room. ρ_c and R_c are *query-independent*, hence precomputed and stored as one scalar per block: the ellipsoidal bound is summary-only at query time too, and the hierarchy is about summary *size* rather than summaries versus keys. That strengthens the reading—at spread 0.02 one extra stored scalar takes the bound from 8.7× to 2.5× the floor, i.e. the implemented bound reads 5.33% of keys against a floor of 2.10%. Measurements of two further refinements show no general gain: seeding the incumbent with precomputed block representatives saves *exactly* zero (branch-and-bound opens blocks in decreasing U_c, so the first block opened already sets $s^\star$ to the true maximum—cost is bound-driven, not incumbent-driven), and an axis-aligned box bound in a shared basis is *worse* than the ellipsoid, with min of the two buying 8% at the most benign geometry alone. The remaining lever is the *partition*.

5.5 The cost of asking: what a summary bound charges on real keys

Everything above counts *keys read*. That is the right metric for Section 5.4, where only the bound varies, and it is the wrong one as soon as the *block size* varies, because evaluating the bound is neither free nor cheap.

An exact summary bound cannot beat a full scan, at any block size. Samuelson's bound needs $q^\top \Sigma_c q$. Held as a dense $d \times d$ matrix that is $O(d^2)$ per block; held as the rank-b centred factor X_c it is $\|X_c q\|^2/b$, i.e. $O(bd)$—*exactly the cost of scoring the block.* Over $B = n/b$ blocks the total is

$$B\,(1+b) \;=\; \tfrac{n}{b}(1+b) \;=\; n + \tfrac{n}{b} \;>\; n \qquad \text{for every } b. \tag{5.8}$$

This is arithmetic and carries no hypothesis about the keys. A search that evaluates the exact summary bound on every block *has already paid for a full pass over the data before it skips anything.* Measured against a scan on real Gemma-2 layer-6 keys at $n = 65536$: $1.062\times$ at $b = 16$ falling to $1.004\times$ at $b = 256$—never below one, and by (5.8) it cannot be.

The routing is excellent; the price of asking is the problem. The same measurement, counting keys alone, has contiguous blocks at $b = 16$ under the exact bound reading **204 of 65536 keys**, a $321\times$ reduction, with k-means at 1519. Real transformer keys *route extremely well.* Every difficulty here is on the cost side of the ledger, not the geometry side.

The escape, and its absence. A rank-r sketch of Σ_c, made admissible by the discarded-eigenvalue tail

$$q^\top \Sigma_c q \;\le\; \sum_{i \le r} \lambda_i \langle v_i, q\rangle^2 \;+\; \lambda_{r+1}\|q\|^2,$$

costs $B(1+r) = \frac{n}{b}(1+r)$, which falls below a scan exactly when $r < b$. So the whole question is one number: how small can r be and still prune? On real keys the answer is that it cannot be smaller than b at all. At $b = 16$ with contiguous blocks, ranks $2, 4, 6, 8, 10, 12$ each read 100.0% of the keys and rank 16 reads 0.3%: a step function, not a gradient.

Why: the block spectrum does not decay. The tail term is exactly zero at $r = b$ and positive below it, so the step is a statement about the spectrum of Σ_c, and it was measured directly. Across $b \in \{16, 32, 64, 128\}$ and both partitions, λ_2/λ_1 lies in $[0.70, 0.84]$—the *first* eigenvalue drop is already small, where a genuinely low-rank block would sit near 0.01— and the participation ratio $(\sum \lambda)^2 / \sum \lambda^2$ never falls below a quarter of the block size, reaching $0.72b$ for contiguous blocks at $b = 16$. Between a quarter and three quarters of all available directions carry real variance. There is no small tail to discard, so no truncation is both cheap and tight.

What this does and does not settle. Together: on real transformer keys, *lossless selection driven by a (μ_c, Σ_c, b) summary costs more than reading every key*, and no rank truncation of that summary escapes it. What it does *not* say is that summarising a block is hopeless—it bounds this family of bounds. Nor does it touch budget-κ *lossy* routing, which is a different object with a different accounting and is what SSA actually ships; the floor of Section 5.4 was always a statement about lossless selection and this is its cost-side companion. The measurements are one model at one layer, and the arithmetic of (5.8) is the only part that is geometry-free.

Distribution-sensitive certified partition objective. Worst-case height is not the only honest unit. For a finite query family with nonnegative weights $w(q)$ and node-read count $r(q)$, define $C(r) = \sum_q w(q)r(q)$. Pointwise fewer reads never increase C, and one strict saving at a query of positive weight strictly decreases it. A safe partition supplies a cell assignment and query-dependent cell bounds dominating every member; only cells whose bounds clear the threshold are read. Lower certified bounds retain a subset and therefore have no larger C. Every supplied finite nonempty family of such partitions has a least-cost member, and the objective equals a tree's weighted node-read cost when their per-query counts agree. This licenses distribution-aware comparison among certified candidates. It does not construct or learn a tree, optimize over all partitions, prove generalization, or turn node count into wall-clock time.

5.6 Sharing one selection across a group of queries

Everything above prunes for *one* query. A long context does not have one query: the number of query positions grows with the context exactly as the number of keys does, and kernels therefore select blocks once for a group—a decode chunk, the query heads of a GQA group, a tile of positions—and read the selected blocks for every member. That is a different object, and it is not automatically lossless: *a bound admissible for one query says nothing about another query's argmax.*

Write Q for the group, $|Q| = m_Q$. The correctness condition is exact and is not a matter of degree. A shared bound U is safe for every member iff it dominates the group's *top claim* $\max_{q \in Q} U_c(q)$ blockwise, and the top claim is the *least* such bound; a chain of prunes that is individually admissible for each stage's own objective can still destroy a later stage's answer, and the part is then gone rather than approximated. These are machine-checked, in `Universal/Potential/ChainedPrune.lean` (`safe_for_every_stage_iff_safe_over_the_top_claim`, `no_safe_bound_holds_below_the_top_claim`, `a_prune_can_drop_the_greatest_part_of_a_second_object`. For the actual pairing claim, $U^*(c) = \max_{q\in Q}\langle q, x_c\rangle$ is the least blockwise function safe for every query. If the shared threshold lies below each query's own greatest pairing, the retained set $\{c : U^*(c) > \theta\}$ contains a maximizing block for every query. Applying one common linear isometry A to all queries and block representatives fixes the claim pointwise because $\langle Aq, Ax_c\rangle = \langle q, x_c\rangle$; the shared safety certificate and retained set are therefore unchanged. This transport result does not cover a different positional map at each query or key slot. Two consequences carry directly into the implementation. The shared *threshold* must be the group's weakest incumbent, not its strongest: a block whose bound has fallen under the best member's incumbent may still hold a weaker member's argmax. And group size costs retention monotonically—each member admitted raises the top claim, and a raised bound prunes strictly less—so in the limit where every block is above threshold for *some* member, the safe prune drops nothing at all.

What it buys, and what it does not. Table 1 counts total work as keys scored plus bound evaluations, since the arms differ in how many bounds they evaluate and counting keys alone credits a saving paid for elsewhere. Sharing the block *choice* at the top claim is worth about 2×, and—the useful part—*flat in group size* up to $m_Q = 512$, because the blocks a tight group wants are the same blocks. As the group spreads, it stops paying and then costs: the quantitative face of the drop-nothing limit above.

q-spread	m_Q	per-query	top claim	rank-8	isotropic
0.02	8	9 014	1.99×	2.31×	1.13×
0.02	32	36 293	2.01×	1.09×	0.63×
0.02	128	142 663	1.97×	0.55×	0.43×
0.02	512	572 668	1.98×	0.18×	0.17×
0.10	32	42 729	0.51×	0.16×	0.16×
0.30	32	112 660	0.43×	0.43×	0.43×

Table 1: Shared selection against per-query selection at the same partition and key bound; speedup on total work. $n = 8192$, $d = 64$, $B = 128$, key spread 0.05. Every arm returns the true argmax for every member.

A negative result: summarising the query side does not pay. The obvious way to make the top claim cheap is to apply this paper's own hierarchy to the queries—the top claim costs $O(m_Q)$ per block, whereas $\max_{q\in Q}\langle q, \mu_c\rangle \le \langle \mu_Q, \mu_c\rangle + \sqrt{(m_Q-1)\,\mu_c^\top \Sigma_Q\, \mu_c}$ costs $O(d)$ once the group's summary is formed. The resulting bounds are admissible and exact, and they are *worse at every group size measured*, degrading from 1.13× to 0.17× as m_Q grows.

The mechanism is structural and is the point worth keeping. Proposition 1 establishes Samuelson's bound as tightest *because it is attained*—by one member at $\bar{x} + s\sqrt{m-1}$ with the rest at $\bar{x}$. That is a worst-case tightness, and a worst case is a statement about the adversarial configuration, not the typical one. On the key side the adversarial block is exactly what a bound must survive. A *tight query group* is the opposite shape: its members cluster near their mean, which is where the $\sqrt{m_Q - 1}$ factor is furthest from the truth. The same inequality that is tight where it was proved is loose where it is reused, and the $O(1)$-per-block evaluation it saves is smaller than the pruning it gives up. The room in shared selection is in the block choice, not in the query summary.

Relation to the formalized ceiling. Maximum-score estimation has a related obstruction in the substrate development: a single score standing for a whole fibre of completions carries an error floor set by the spread of the objective across that fibre (`Universal/Potential/PartialScore.lean`, `score_error_ge_of_reach_split` and `not_exactOnReach_of_reach_split`). Here the partial object is the summary, its completions are the blocks carrying it, and the objective is the block's true maximum logit. The instantiation is stated in prose and is *not* machine-checked; see `docs/substrate_math_imports.md` for the mapping and its limits. This score-approximation floor is distinct from the attained-upper-bound argument of Proposition 1; variation among individual logits in a fixed block does not by itself preclude exact identification of that block's maximum or correct block selection.

The abstract skeleton is machine-checked, axiom-pure, in `Universal/Potential/Admissible-Bound.lean`: a pointwise tighter bound provably drops a superset of the parts (`a_higher_bound_drops_a_subset`, `dropped_card_mono`)—the qualitative claim of Section 5.1 as a theorem; the family maximum is the least admissible bound (`familyMax_le_of_isAdmissible`); an attained bound admits nothing smaller (`no_bound_below_an_attained_one`), which is Proposition 1 in abstract form; and the minimum of two admissible bounds is admissible (`min_isAdmissible`), which is what licenses the implementation's min of the isotropic and ellipsoidal bounds. Finally, `at_the_floor_a_maximal_threshold_drops_every_part` predicts the null above: at the floor a maximal threshold drops every part, so a search still reading parts is paying for bounds above the floor and a better incumbent cannot help it.

5.7 Certifying omitted mass and attention-output error

An exact argmax or routing-metric top-κ certificate does not control the total softmax weight of omitted keys. In particular, many individually small logits can carry most of the partition sum. For a nonempty kept set S, let $D = S^c$ within the visible causal prefix, and define

$$Z_S = \sum_{i \in S} e^{\beta a_i}, \qquad Z_D = \sum_{i \in D} e^{\beta a_i}, \qquad \widehat{o} = Z_S^{-1} \sum_{i \in S} e^{\beta a_i} v_i.$$

Suppose D is partitioned into unopened blocks. Each block has size b_c, key mean μ_c, key radius R_c, value mean ν_c, and value radius r_c^V. For $\beta \geq 0$ set

$$L_c = b_c e^{\beta\langle q,\mu_c\rangle}, \qquad A_c = b_c e^{\beta(\langle q,\mu_c\rangle + \|q\| R_c)}, \qquad L = \sum_{c \subseteq D} L_c, \quad A = \sum_{c \subseteq D} A_c. \tag{5.9}$$

Jensen's inequality and the admissible radius bound give $L_c \leq Z_c \leq A_c$ without scoring the unopened keys. Other admissible maximum-logit bounds can replace the radius term, with their evaluation cost charged separately.

Proposition 2 (Subset-attention certificate)**.** *Let p be the dense softmax distribution and $\widehat{p}$ its renormalized restriction to S, extended by zero outside S. With $\delta = Z_D/(Z_S + Z_D)$,*

$$\mathrm{TV}(\widehat{p}, p) = \delta \leq \overline{\delta} := \frac{A}{Z_S + A}, \tag{5.10}$$

$$\mathrm{KL}(\widehat{p}\|p) = -\log(1-\delta) \leq \log(1 + A/Z_S). \tag{5.11}$$

For $d_c = \|\nu_c - \widehat{o}\| + r_c^V$ and nonempty D, the output obeys

$$\|o - \widehat{o}\| \leq \min\left\{\overline{\delta}\max_{c \subseteq D} d_c, \quad \frac{\sum_{c \subseteq D} A_c d_c}{Z_S + L}\right\}. \tag{5.12}$$

All three bounds are zero when D is empty.

Proof. On S, $\widehat{p}_i = p_i/(1-\delta)$, so the absolute weight difference sums to δ on each of S and D. The likelihood ratio on S is constant, giving the KL identity. Both expressions increase with Z_D, which is at most A. For the value bound, subtract $\widehat{o}$ from the full weighted average:

$$o - \widehat{o} = \frac{\sum_{i \in D} e^{\beta a_i}(v_i - \widehat{o})}{Z_S + Z_D}.$$

Each omitted value in c lies within d_c of $\widehat{o}$. Bounding all distances by their maximum gives the first term; bounding each block numerator above by $A_c d_c$ and the denominator below by $Z_S + L$ gives the second. $\square$

The direction of KL is essential: for finite logits and a proper restriction, $\mathrm{KL}(p\|\widehat{p}) = +\infty$. At equal logits, $\delta = 1 - |S|/n$ and $\mathrm{KL}(\widehat{p}\|p) = \log(n/|S|)$, the nested-uniform-support identity formalized by `klDiv_uniformSupport` in `Universal/Potential/Entropy/UniformSupport.lean`. The finite-vector Pinsker theorem `totalVariation_le_sqrt_klDiv` in the same directory's `TotalVariation.lean` allows zeros in its first distribution and a strictly positive second distribution; it applies in this direction. Equation (5.10) directly computes the TV quantity for a restriction, so using Pinsker would give a weaker bound. The partition bound consumes the reasoning of `logSumExp_max_sandwich`. The complete abstract proposition is now machine-checked by `RestrictedReadBound.lean` and `RestrictedReadOutputBound.lean`: they include the exact TV and KL identities, the exact output residual, both block-certificate arms and their minimum, the equal-value fence, and the reverse-direction unbounded limit. The SSA summary construction and Python implementation are not Lean proofs.

One potential and an append-only store. The finite attention surface can be specified by only scores $a_i(q)$ and stored values v_i. Its potential $\Phi(q) = \log \sum_i e^{a_i(q)}$, normalized weights, dense read, and selected read are then derived rather than independent objects. For scalar values, tilting every score by tv_i gives

$$\frac{d}{dt} \log \sum_i e^{a_i(q)+tv_i}\bigg|_{t=0} = \sum_i \frac{e^{a_i(q)}}{\sum_j e^{a_j(q)}} v_i. \tag{5.13}$$

This identifies the Hopfield/attention read with the directional derivative of the same log-partition whose first two score moments drive cumulant routing. It does not identify the two-moment approximation with the potential.

The distinction between stored content and access weight is exact for an append-only store. Let a nonempty old store have exponential mass Z, normalized read r, and append a fresh value v with exponential mass $a > 0$, holding the score surface fixed. Then

$$Z' = Z + a, \qquad \Phi' - \Phi = \log(1 + a/Z), \qquad r' - r = \frac{a}{Z+a}(v - r), \tag{5.14}$$

and every old normalized weight is multiplied by $Z/(Z + a)$ and therefore strictly falls. If scores also change between rounds, both potential change and read change split exactly into the fresh-site term evaluated at the new scores plus a score-only term on the unchanged old store. No degradation direction follows: the fresh value can equal r, and a score change can reinforce or cancel the append term. Nothing is deleted or rewritten, and no cumulative drift, convergence, or recovery law is implied.

Sequential routed reads. Local output certificates do not simply add through a transformer unless the intervening exact maps control how state error propagates. Let F_t be the exact step, G_t its routed approximation, $\|G_t(x) - F_t(x)\| \leq \epsilon_t$, and let F_t be L_t-Lipschitz. For exact and approximate paths starting from the same state,

$$e_{t+1} \leq \epsilon_t + L_t e_t, \qquad e_0 = 0. \tag{5.15}$$

Iterating this recurrence is the certified multi-hop bound: an early error is multiplied by every later gain. If every $L_t \leq 1$, then $e_N \leq \sum_{t<N} \epsilon_t$; exact local steps give identical paths. The gain hypothesis is necessary: a two-step scalar construction can make the first local error exactly one and the second local error zero while a shared, arbitrarily high-gain second map makes the final error exceed any proposed bound. This is deterministic state-error composition, not

multiplication of retrieval success probabilities, and the cumulant score interval alone supplies neither ϵ_t nor L_t.

Adaptive implementation and cost. `ssa/certified_attention.py` builds immutable contiguous-block key/value summaries and opens blocks in decreasing $\log A_c$, optionally starting with blocks selected by another router. It evaluates attention exactly on opened keys, checks every requested mass or output tolerance, and doubles the number of opened blocks after a failed check. A block cap returns an explicitly uncertified result if the tolerance remains unmet. A partially visible causal block is opened using only its visible keys; its full-block summary is excluded from the decision. Log-space partition sums and suffix reductions avoid exponential overflow and subtraction of nearly equal masses. Bounds are evaluated in float64 with a small outward score cushion; this is not an interval-arithmetic guarantee.

Construction costs $O(n(d + d_v))$. For B blocks, each query pays $O(Bd)$ for key bounds, $O(B \log B)$ for ordering, $O(Bd_v \log B)$ for value-certificate checks in the worst case, and $O(|S|(d+d_v))$ for opened keys and values. The reference reports key scores, key-bound evaluations, certificate checks, and value-bound evaluations separately. It can scan all keys on diffuse geometry and does not establish a sublinear per-query cost at fixed block size or a GPU speedup. The exact covariance factor whose cost is analyzed in Section 5.5 is not evaluated by this radius-based reference.

geometry	keys scored	$\overline{\delta}$	output bound	measured error
concentrated	64	9.54×10^{-6}	4.62×10^{-5}	1.79×10^{-6}
flat logits	4096	0	0	8.58×10^{-17}
equal values, flat logits	64	0.984375	0	0

The deterministic CPU fixture uses $n = 4096$, $d = 32$, $d_v = 8$, $b = 64$, $\beta = 4$, seed zero. The first two rows request omitted mass at most 10^{-3}; the last requests only output error at most 10^{-12}. Every row evaluates 64 key bounds. Certificate checks number $1, 7, 1$, and value-bound evaluations number $63, 321, 63$, respectively. The full-scan residual is floating-point roundoff. Equal values permit exact output even when almost all attention mass is omitted; conversely, a small omitted weight can matter when its value is sufficiently distinct. These are controlled mechanism measurements, reproducible with `python -m ssa.certified_attention`.

Variance-sensitive mass refinement. The worst-radius mass cap can be sharpened without changing the target attention distribution. Let a node contain m keys, let $x_i = \langle q, k_i - \mu \rangle$, suppose $x_i \le R$, and write $\sigma^2 = m^{-1} \sum_i x_i^2$. For $R > 0$ and $\beta \ge 0$,

$$\sum_i e^{\beta \langle q, k_i \rangle} \le m e^{\beta \langle q, \mu \rangle} \left[1 + \frac{\sigma^2}{R^2} \left(e^{\beta R} - 1 - \beta R \right) \right]. \tag{5.16}$$

If $R = 0$, equality holds. To prove (5.16), apply $e^{\beta x} \le 1 + \beta x + (x^2/R^2)(e^{\beta R} - 1 - \beta R)$ for $x \le R$, sum, and use $\sum_i x_i = 0$. For vector keys, $R = \|q\|R_c$ and $\sigma^2 \le \|q\|^2 s_c^2$, where $s_c^2 = m^{-1} \sum_i \|k_i - \mu_c\|^2$. This trace lift is universally safe; diagonal coordinate variances alone do not determine covariance, so the assumption-free diagonal alternative pays a dimension factor. If the node instead stores its full covariance C_c, then $\sigma^2 = q^T C_c q$ exactly: expanding the quadratic form gives $\sum_i \langle q, k_i - \mu_c \rangle^2 = m q^T C_c q$. This removes the dimension factor at the explicit cost of $O(d^2)$ storage per node and a quadratic-form evaluation per query/node.

The `bennett` and `bennett_covariance` modes store these two summaries recursively, evaluate (5.16) stably in log space, and take its minimum with the radius cap. On an 8,192-key concentrated synthetic control, median excess above exact node log mass fell from 0.816 to 0.0081 (trace) and 0.000026 (covariance); either mode cut a 10% omitted-mass certificate from 3,884 to 556 scored keys. The real Qwen control is negative: across 8,160 query/node pairs from 32 causal queries in an 8K post-RoPE layer-18 head, trace cut the cap by a median 0.730 log units and covariance by 6.057, with no observed float64-oracle underestimate. Yet median excess remained 33.42 and 28.03 log units respectively, and all modes opened every visible block to certify either 10% or 1% omitted mass. Full covariance therefore isolates the remaining obstruction as the worst-radius exponential tail, not the trace lift's dimension loss. Variance-sensitive mass is a real benign-geometry improvement, not the missing sparse exact certificate on this raw model geometry. The record is `runs/bennett_mass_tree.json`; reproduce it with `python -m ssa.bennett_mass_experiment` on a CUDA host with cached Qwen weights.

A deterministic outlier peel attacks that tail directly: expose the t largest $\|k_i - \mu\|^2$ values with ordered tie-breaking, score their mass exactly, and apply (5.16) to the recentred core. If the original trace spread is s^2, every unexposed residual has squared norm at most $ms^2/(t + 1)$, and its distance from the new core mean is at most $2\sqrt{ms^2/(t + 1)}$. Exact exposed mass plus any valid core cap is valid; a running minimum over peel depths is valid and monotone even though the raw caps need not be. On the same Qwen fixture, trace peels $t = 1, 2, 4, 8$ leave median slack 29.83, 29.33, 28.80, and 27.92 log units. Four peeled vectors plus core covariance reaches 23.39, but every variant still opens every visible block. At $d = 64$, the latter stores 4,418 extra scalars per node. Small peeling is therefore a large tightening, but not a sparse exact certificate; large peeling converges toward storing and scoring the keys. Force-keeping the exposed vectors through their containing leaf blocks does not change that conclusion: at $t = 4$ it seeds 8.34 blocks and reduces mean bound evaluations from 190.63 to 166.78, but still reads all 98.16 blocks and 6,247.25 keys at both tested stopping tolerances. This is a block-kernel realization, not an individual-key outlier side channel.

Six additional checks compare the certificates against float64 CUDA SDPA on an RTX 4080, using concentrated logits, flat logits, and equal values at $n = 1024$, $d = 32$, $d_v = 8$, $b = 32$, with visible prefixes of 1024 and 997 keys. These validate numerical agreement with an independent attention implementation, including partial causal blocks. The selector runs on CPU; GPU routing speed and real-model quality remain unmeasured.

6 The trilemma and the grounded-probe limit

Call a selector *cheap* if it reads $o(n)$ keys, *lossless* if it attends every key dense attention would weight non-negligibly, and *length-robust* if its accuracy is flat in n. A precise probe-model limit is:

Proposition 3 (Bounded grounded reads miss)**.** *Consider an adaptive binary decision tree over n positions, where probing position i reports whether the unique spike is there. Suppose every returned position was actually probed on that input. If every root-to-return path has length at most b, then the selector recalls the spike on at most b of the n placements, hence has recall at most b/n under a uniform placement. For any finite randomized mixture of such selectors, some one fixed placement has seed-averaged recall at most b/n.*

Proof. Follow the all-false reference run and call its probe trace E; $|E| \le b$. If a spike is planted at $j \notin E$, every answer along that path remains false, so the entire trace and returned set are unchanged. Because returned positions must lie in that trace, this run cannot return j. Thus every recalled placement lies in E, proving the deterministic count and uniform rate. For a finite mixture, sum each seed's recall indicators over placements, average over seeds, and exchange the two finite sums. The total is at most b, so at least one placement has seed-averaged recall at most b/n. □

The grounded-output hypothesis is essential: a depth-zero procedure can return the whole carrier and thereby "recall" every spike without probing one. This theorem by itself does not cover a preprocessed index whose stored side information can name unprobed positions; the finite-state extension below covers one explicit capacity model, while an arbitrary index still requires a budget for its information. For grounded adaptive probes, however, worst-case losslessness forces $b \ge n$. `BoundedReadMiss.lean` machine-checks the reference-run invariance, deterministic count/rate, finite-randomized extension, sharp fixed-position reader, and the zero-depth counterexample.

A finite-state abstraction now makes part of the preprocessing boundary quantitative. Suppose the planted position selects one of K index states, each state selects an adaptive read of depth at most b, and its all-false reference leaf may additionally return at most a unprobed positions. The reachable placements are covered by the K statewise reach sets, so

$$\#\{\text{recalled placements}\} \le K(b+a). \tag{6.1}$$

For a finite randomized family, exchanging the seed and placement sums shows that some fixed placement has seed-averaged recall at most $K(b+a)/n$ and miss share at least $1 - K(b+a)/n$. If every statewise output is grounded, the unread-output allowance disappears: the deterministic ceiling is Kb, and the same averaging argument gives Kb/n. The capacity factor is necessary: the identity index uses $K = n$ states, records the spike position, and returns its singleton at depth zero, recalling every placement. This is finite counting, not a general data-structure lower bound: K counts states but does not charge their bit representation, index construction, state lookup, arithmetic between probes, or scored/approximate/multiple target retrieval.

Proposition 3 says worst-case losslessness forces a full read *in its grounded probe model.* A quantitative companion for attention holds under fine-grained complexity assumptions (SETH): even *approximating* the attention output requires $n^{2-o(1)}$ time once entries reach $\omega(\sqrt{\log n})$, and the truly subquadratic algorithm that exists in the bounded-entry regime is itself an approximation (a low-rank polynomial method), not exact computation [14]. No unrestricted indexing lower bound is inferred from the probe theorem. The three-way reading below is an organizing *taxonomy* whose third axis is measured in Sections 7–10, not a proved trichotomy: operationally *two* of {cheap, lossless, length-robust} are available at once:

keep	give up	what it is
lossless + length-robust	cheap	dense attention—reads every key
cheap + length-robust	lossless	approximate selection—fast, flat, can miss a worst-case spike
cheap + lossless	length-robust	branch-and-bound on *benign* keys—bounds tight, low cost

Note the third row concedes length-robustness of *cost*, not of accuracy: admissible branch-and-bound never returns a wrong answer—off benign geometry it degrades by *reading more* (toward everything), whereas row 2 trades accuracy. "Length-robust" in the first two rows is the accuracy sense of Section 3; keeping the two currencies straight is part of the taxonomy's honesty.

The escape from the trilemma is the third row: cheapness *and* losslessness are jointly available *when the bounds* (5.1)/(5.2)/(5.6) *are tight*, which is a property of the key geometry, not of the algorithm. SSA is therefore not claimed as universal subquadratic exact attention, but as a mechanism that is cheap, lossless, *and* length-robust *on benign geometry*, and merely cheap-and-length-robust-but-approximate otherwise. The next section is about making the geometry benign.

7 Manufacturing routability

7.1 Benign geometry, precisely

The branch-and-bound cost and the prune gate (5.6) are governed by the *off-target spread* $q^\top \Sigma_c q$ for the blocks a query does *not* need, relative to the *margin* Δ to the block it does. Geometry is *benign* for a query q when, for every non-target block c,

$$\langle q, \mu_c \rangle + \sqrt{(b-1)\, q^\top \Sigma_c q} \;<\; s^\star(q), \tag{7.1}$$

i.e. the prune gate (5.6) fires everywhere except the target's block. Then exactly one block (plus the local window) is opened, branch-and-bound reads $O(b)$ keys, and selection is cheap *and* lossless. Isotropic keys violate (7.1)—every block's bound is large and uninformative, so nothing prunes and cost reverts to dense. Condition (7.1) is *sufficient*; the matching necessary-side statement is Proposition 1 and the floor measured beneath it in Section 5.4, which bounds how much benign geometry can ever buy.

7.2 The routability regularizer

Benign geometry is not given; it can be *trained in.* Add to the language-model loss a penalty that shrinks the off-target spread along the directions queries actually use:

$$\mathcal{L} \;=\; \mathcal{L}_{\mathrm{LM}} \;+\; \lambda \sum_i \sum_{c \neq \mathrm{target}(i)} q_i^\top \Sigma_c\, q_i. \tag{7.2}$$

Minimizing $\sum_{c \neq \mathrm{target}} q^\top \Sigma_c q$ is exactly minimizing the quantity that appears under the square root in the prune gate (5.6), so as training proceeds the gate fires for more non-target blocks and the certified bound (5.1)/(5.2) tightens. Measured: co-training with (7.2) drove the *lossless* branch-and-bound cost from 26.5% of keys to 4.2%—a 6× reduction—at *no* loss of retrieval accuracy. Training is what manufactures the geometry that the impossibility result (Proposition 3) says cheap exactness requires. It does so against a floor: Section 5.4 measures the summary price falling from 63× to 8× the partition's own limit as spread tightens, monotonically and without reaching it.

7.3 The entry-magnitude split: selection vs. linear attention

A complementary fact decides *which* subquadratic route a given layer should take. Let B be the characteristic magnitude of the logits $\langle q, k \rangle$ (the "entry scale"). The exponentiated score matrix $[\, e^{B\langle q_i, k_j \rangle}\,]$ has effective rank that *grows with* B: for small B it is nearly low-rank (so a kernel/linear-attention factorization $\phi(q)^\top \phi(k)$ is accurate and $O(n)$); for large B it is full-rank (so no linear map approximates it, and one must *select*). Empirically, the effective rank of $e^{BKK^\top}$ on 256 keys rose from 2.7 at $B = 0.5$ to 256 (full) by $B = 8$.

The selection route, by contrast, is *scale-invariant*: in the prune gate (5.6) both the squared margin and the spread scale as B^2, so the gate's truth value is unchanged by B. Selection therefore works at *any* entry scale, whereas linear attention works only in the small-B (smooth) regime. Sharp, long-range retrieval is the large-B regime—which is why selection, not linearization, is the route for long-context exact recall.

8 Length generalization and staged extension

A selector that is flat in n (Section 3) still needs the *model* to behave at lengths beyond those it was trained on. The position encoding decides whether it can.

8.1 Position-invariant routing under rotary embeddings

With rotary position embeddings (RoPE), a query/key pair interacts only through their *relative* offset: the algebraic score depends on $i - j$, not on i, j absolutely. This removes an absolute-position table, but it does *not* prove that a trained model transfers to arbitrary unseen offsets. Each rotary band is periodic. If its per-position phase advance is bounded by ω, a nonzero wrapped turn over a window requires at least one wavelength, $N \geq 2\pi/\omega$; below that length the winding is zero. Moreover, two sampled phase schedules that differ by less than half a turn at every position have the same winding, so changing the turn count forces that anti-aliasing margin to fail somewhere. Extending far enough is therefore a change of winding regime, not merely a harmless translation of a relative coordinate.

Content-only routing can still transfer empirically because its proposal geometry omits the rotary action, while the attention score keeps the model's post-RoPE vectors. The measured toy staging result and the real model's roughly 2–4× zero-shot range support limited transfer; they do not establish an unbounded RoPE theorem. This distinction is especially important for the 10M run's approximately 306× static YaRN factor.

8.2 The staging ladder

Zero-shot extrapolation buys a factor, not an unbounded range. To go further, *stage*: extend the context, briefly continue-train (adapt) at the new length, extend again—doubling each rung. The economics are favorable precisely because routing is position-invariant: each rung extrapolates only 2× from the last *adapted* length, so every adaptation starts from the $\sim$2× zero-shot recall and only has to clean it up. The adaptation cost is small and roughly *flat in length* rather than growing.

Measured (a 6-layer model on a synthetic multi-query associative-recall task; base trained at 48 pairs):

rung	multiple	zero-shot recall	adapt steps	recall after adapt	recall with SSA
96	2×	0.993	100	1.000	1.000
192	4×	0.904	100	0.999	1.000
384	8×	0.810	100	1.000	0.995
768	16×	0.672	100	0.996	0.996
1536	32×	0.516	400	0.982	0.979

The base schedule cost 7600 steps; the *entire* climb from 16× (where zero-shot recall is already near the floor) to 32× at recall 0.982 cost only 800 additional steps—about 10% of the base, and roughly constant per rung. The final column confirms that the same adapted model runs under SSA selection ($\leq$ 15% of keys attended) with essentially no recall loss at every rung.

9 The construction pipeline: dense to subquadratic

SSA can be retrofitted onto an existing dense pretrained model, which is how a frontier long-context system is built without pretraining from scratch:

1. **Start** from a dense pretrained base.
2. **Swap** dense attention for SSA (Algorithm 1) in every layer.
3. The swap **degrades** quality, because the base was trained expecting every key.
4. **Adapt** by continued pretraining; the keys co-adapt to the sparse routing (and, with the regularizer (7.2), become routable).
5. **Stage** the context extension (Section 8).

Measured on a 124M-parameter dense base, held-out perplexity (lower is better):

condition	perplexity
dense base (off-domain start)	32.5
+ SSA swap, no adaptation	45.8
dense base + equal in-domain continued training (control)	23.9
SSA-swapped + equal in-domain continued training	**25.1**

The swap costs +13.2 perplexity; an *equal-budget* adaptation recovers 94% of that gap, landing within +1.2 of a dense model given the *same* continued training—while attending only ∼38% of keys at this configuration. The control matters: continued training lowers perplexity for either attention (domain adaptation), so "recovery" must be measured against a dense model given the same steps, not against the off-domain start. The residual is the price of sparsity at this small budget and shrinks as the budget grows.

10 The compute floor and a sub-linear router

Selection caps the per-query work at a budget κ keys, so the irreducible cost of the layer is the *attention floor* $n\kappa$—linear in n. A measured kernel sits above this floor by exactly its router cost. Decomposing one forward of the kernel of §9 into router (the $(n/b)^2$ block-score GEMM), `BlockMask` construction, and attention, and fitting each over $n \in [16\text{K}, 524\text{K}]$, gives

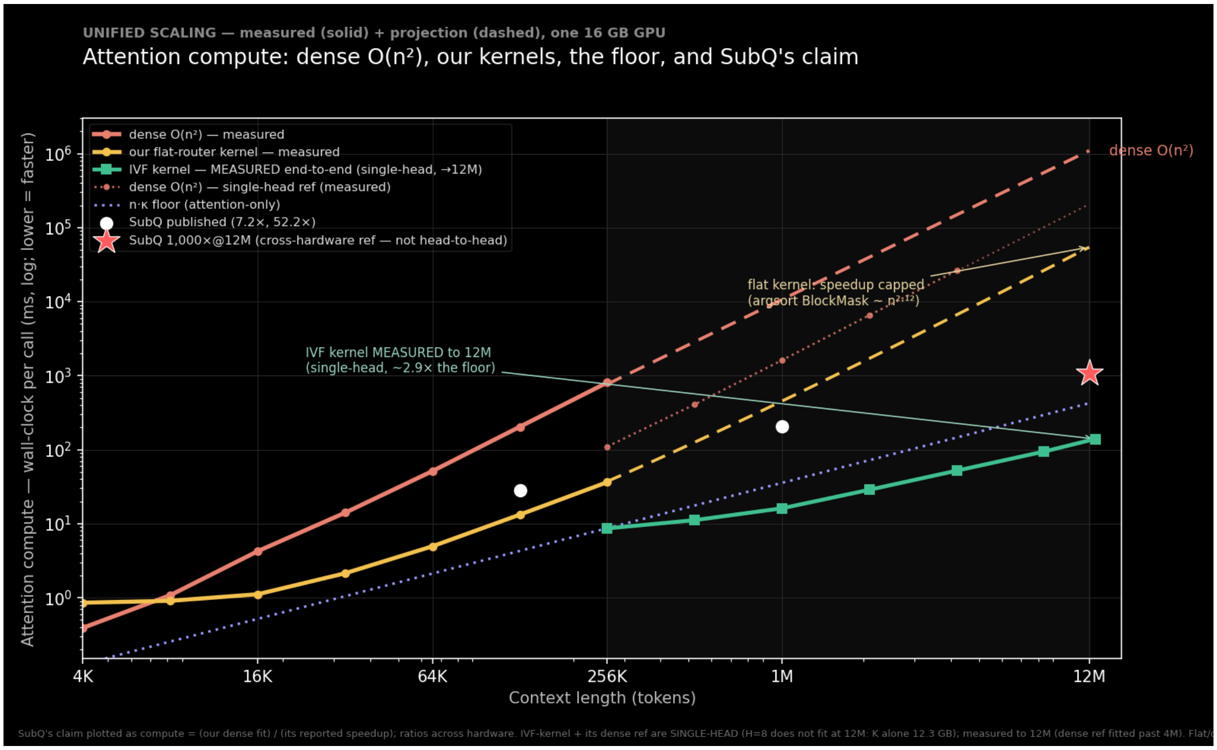

Figure 1: Attention compute versus context length (lower is faster). Dense $O(n^2)$ rises at the top; our flat-router kernel (measured) stays well below it but its speedup is capped because the argsort `BlockMask` build grows nearly as fast; the measured faiss-GPU IVF router drops the kernel onto the $n\kappa$ floor. SubQ's two published speedups and its 1,000×@12M claim are shown as compute = dense/speedup. Measured solid, projection dashed; one 16 GB GPU.

attention $\sim n^{1.02}$ (the floor), router $\sim n^{1.76}$, and—largest—the argsort-based mask build $\sim n^{2.12}$. Extrapolated to 12M tokens the floor is ~1% of the forward: a 128× gap, entirely the two $(n/b)^2$ terms. A router that emits the selected block indices *directly* removes both (Figure 1).

Lowering the floor. The floor $n\kappa$ itself drops if fewer keys recover the target. The smallest budget $\kappa_{\min}$ reaching recall ≥ 0.9 is $\kappa_{\min}/n = 3\%$ for tight clusters, 50% for diffuse or adversarial geometry (no compression), and the routability regularizer of §7 drives it from 25% ($\lambda = 0$) to **0**.4% ($\lambda = 64$)—a 60× reduction, the dominant lever, on benign geometry only.

Closing the gap. The necessity argument of §6 forces a sub-linear, examine-$o(B)$ index. A bake-off on benign co-trained keys (recall vs. keys scored) ranks an IVF (inverted-file) router and a recursive-radius treecode ahead of LSH, which fails to reach recall 0.9. An IVF over the block means scores $O(\sqrt{n/b})$ blocks at 0.93–0.97 block-selection agreement with the flat router and emits `kv_idx` directly. Timed on a single 16 GB GPU with both routers on-device (no host transfer), the dense $(n/b)^2$ GEMM's constant wins below ~2M but its score matrix exhausts memory at 4M; the IVF runs linearly past it:

context n	flat $(n/b)^2$ GEMM	faiss-GPU IVF	winner
256K	0.21 ms	7.3 ms	flat 35×
2M	14.0 ms	19.5 ms	flat 1.4×
4M	52.6 ms (runs)	31.4 ms	IVF 1.7×
8M	OOM (nb^2=17.2 GB)	63.7 ms	IVF only

The crossover is ∼3M. This stripped single-head GEMM OOMs only at 8M (17 GB matrix), while the kernel's *actual* router (`block_route`, H heads + an argsort) OOMs near 1M—so the IVF is the only router that runs at long context and is measured to 12M (94 ms).

The gap, closed end-to-end (measured). Wiring the IVF router into the kernel—it emits the compressed block-index contract directly, and the mask is built without the dense $(n_b, n_b{+}1)$ transpose (which would need 38.7 GB at n_b=98,304)—lets the whole forward be measured, single-head, to 12M on one 16 GB GPU: router 101 ms, mask build 0.003 ms, attention (the floor) 47.5 ms, *total* 139.5 *ms* in 6.55 GB. The direct-index mask build is 0.003 ms, compared with a 40.7 s extrapolation for the argsort path, and the residual gap to the floor is a measured 2.9×. The scope is single-head (H=8 does not fit at 12M) and synthetic keys (a speed result), and the whole result is conditional on benign geometry—adversarial or multi-hop retrieval returns $\kappa_{\min}$ to the 50% floor, where no speedup exists. Both ingredients of a quality-preserving large-context speedup—a floor-lowering training stage and a sub-linear indexer—are thus exhibited, and the indexer is shown driving a live kernel to ∼the floor, under exactly that benign-geometry condition.

11 Experiments

The evidence below is organized by current claim and fixture. Synthetic timing establishes kernel scaling; controlled training establishes mechanism behavior; frozen pretrained models establish integration and retrieval behavior. Only the complete 10M experiment combines real-model geometry, every layer and head, long context, subquadratic routing, sparse execution, and an observed quality outcome.

Routing. Second-order (cumulant) routing recovers targets where centroid routing collapses, matching the $1/b$ outlier-attenuation analysis of Section 5.2; routing quality peaks near $\beta \approx 2$, consistent with the bias–variance reading of (5.3).

Kernel speedup. A block-sparse implementation of Algorithm 1 achieved a 20.6× wall-clock speedup over a dense exact kernel at $n = 262{,}144$ on a single accelerator, with the crossover well below that length.

The IVF kernel, measured to 12M. Wiring the sub-linear IVF router into the kernel (Section 10) and measuring the whole forward single-head on one 16 GB GPU runs a 12*M-token* *forward in* 139 *ms and* 6.55 *GB*, at a measured 2.9× the $n \cdot \kappa$ floor (versus the 128× gap the flat kernel projected); the argsort mask build collapses to sub-millisecond because the IVF emits block indices directly. The autoregressive decode step is *flat in* n (∼0.6 ms from 1M to 12M at fixed κ) while a fair fp16 flash-decode step's prefix read grows with n (0.5 → 5.3 ms)—a 9× per-step gap at 12M with the crossover near 1M–2M, both measured. The comparison uses the fair fp16 flash-decode row; a naive fp32-upcasting reference remains in the benchmark only as a diagnostic. This is a single-head synthetic-key speed result; selection quality is a separate benign-geometry claim.

Multi-hop composition. A chained retrieval through the same budgeted block selector obeys the composition law chain $\approx \prod_j \rho_j$: benign single needles hold at 1.00 while a *mixed* two-hop chain (one benign hop, one isolated hop) collapses to 0.02 at $n = 65{,}536$—the isolated hop's $\rho \approx 0.02$ divides the product. The multi-hop sag is the single-needle result read h times, reproducing the NIAH≫MRCR benchmark split as a prediction of the same theory rather than an anomaly.

The fused kernel inside a real model. Swapping the kernel into a pretrained Qwen2.5-0.5B and measuring at 8K–128K preserves single-needle NIAH at 1.00 while giving a 1.5–1.6× prefill speedup at 32K (budget 0.06–0.12), the speedup growing with n and with a tighter budget—the synthetic-key crossover shape inside a real model. At matched budget the analytic $O(n^2)$ path needs 10.7 GB and 3.5 s where the fused kernel needs 1.4 GB and 130 ms, and the analytic path OOMs before 64K while the kernel reaches 128K (under YaRN). This configuration jointly measures a real model, long context, a subquadratic kernel, and quality at 0.5B scale.

Complete pretrained transformer beyond 10M. The memory-bounded Qwen2.5-0.5B path processes 10,000,128 *tokens* through all 24 layers, 14 query heads, and 2 KV heads on one RTX Pro 6000. It completes in 713.2 *s* at 14,022 *token/s* with 25.72 *GB* peak CUDA allocation. The fixed-beam center-radius tree routes on pre-RoPE content geometry; native GQA sparse attention scores the selected blocks with post-RoPE Q/K. Layer-1 cross-head consensus and a bounded 128-block cross-layer reservoir retain route evidence while keeping the per-query/head budget at no more than 198 blocks, an upper selected fraction of 0.506%. The 4K dense-equivalence gate, dense and streamed 128K NIAH gates, and the 10M semantic ranking pass (`walnut` 5.844, best distractor 4.438). The model is frozen and uses static YaRN at approximately 306× its training range. Consequently this experiment demonstrates the full execution conjunction and one successful quality outcome, not dense-equivalent output or general long-context quality.

An optimal selector: the Certified Causal Cascade. Composing five ingredients—a shared low-dim routing space, sub-block max-pool summaries, a chunked-causal streaming index, an exact outlier side-channel, and per-query admissible certificates with escalation—into one streaming selector. The certificate is sound (certified $\Rightarrow$ the selected blocks equal the parent-index-tie-broken top-κ under the routing metric; zero violations on clustered and random geometry). In the indexed regime its skip and truncation checks are strict at the returned threshold; an exhaustive tie instead follows the explicit index policy. This certifies the routing metric, not the omitted softmax mass or attention-output error; the latter have a separate reference certificate in Section 5.7. The component table is the trilemma made concrete: sub-block granularity and the outlier channel rescue high-norm spikes, but *isolated unit-norm needles stayed unretrievable for every tested cheap selector* (recall 0.05)—the grounded-probe obstruction of Section 6 in miniature, not an unrestricted indexing lower bound. Per-layer routing is ∼59% of a Qwen-0.5B prefill (at DSA's reported 58%), and the lever that makes it cheap is **cross-layer sharing from a mid donor layer**—measured cutting it to ∼6% with single-needle retrieval preserved, consistent with the analytic ÷5 estimate; sharing from layer 0 fails. A trained d_r=16 projection reaches 0.65 block agreement on real keys versus 0.32 untrained, but is itself too lossy to drive retrieval—the honest boundary.

The compression corner, measured. The trilemma has a second corner—a fixed- or growing-state memory written at inference time (the "zero attention" / DeltaNet/Titans family). Small exact reference memories, measured against Lean predictions, place it. The READ rule sets the capacity class: a linear read $o = Sq$ is rank-d capped (recall collapses at $m \approx d$; `read_capacity_le_dim` / `rank_d_read_wall` prove the $\leq d$ ceiling) while a softmax read over the same pairs holds far past $m = d$ (measured to $m = 512$; `softmax_capacity` gives the exponential form)—capacity is a property of the read, not the substrate, with machine-checked results on both sides of the contrast. The load-bearing measurement (empirical, no theorem) is that **compression $\neq$ selection**: a needle salient only at read time is lost by a surprise-gated fixed memory (recall 0.10) where selection recovers it (1.00)—write-time compression cannot keep what the future query has not yet made relevant. A distribution shift is a fold a fixed memory cannot track (`fold_not_hopfield`: pre-shift recall decays 0.90 $\to$ 0.10; a growing slot-birth state holds 0.65), and the proved composition bound $\prod \rho \leq$ min-hop (`chain_le_weakest`) is reproduced, the measured joint chain sagging below $\prod \rho$. The NIAH-$\gg$-multi-hop split is thus architecture-independent —it holds whichever corner of the trilemma one builds.

The compression corner, trained. The reference memories above are untrained; the zero-attention recipe's load-bearing half is *training-dependent*—a learned write gate and an auxiliary future-prediction objective. The trained fixture is a small micro-LM ($d = 128$, head dimension $d_h = 16$) trained end-to-end on MQAR with a token-mixer swappable between the two corners at matched state (DeltaNet state d_h vs. an SSA budget $\kappa \approx d_h$). Three measurements. (i) *Capacity:* trained selection (dense, SSA) is flat in load, while the trained DeltaNet groks the task and holds to $m \approx d_h$ then walls at the same rank-d_h boundary—training moves the wall, it does not remove it. (ii) *The learned write gate has no measured benefit:* on write-salient MQAR (keep-worthy pairs use reserved marker keys, identifiable at write time) the no-gate delta rule already solves it—training shapes the $\leq d_h$ keepable keys itself; on read-salient MQAR nothing lifts the compression wall, gate or no gate. (iii) *The future-prediction auxiliary loss is flat in its weight* on the read-salient wall. So the training-dependent half of the recipe does not close the gap: where keeping is possible training already does it, and where relevance is read-time-only no write policy can serve it—the trained mirror of the 0.10-vs-1.00 split, and the composition sag persists for the compression corner under training as well. The bottom line: dropping attention does not dissolve the trilemma but relocates within it—a compression memory *works* where relevance is fixed at write time and within its state (write-salient recall, NIAH), and *fails*, by capacity rather than by any trainable objective, on read-time-only relevance, past-capacity retrieval, and multi-hop chains. This is why the frontier long-context state-space models remain *hybrids* with interleaved attention.

Routability. The regularizer (7.2) reduced lossless branch-and-bound selection cost from 26.5% to 4.2% of keys. (Accuracy is 1.000 by construction—admissible-bound branch-and-bound always returns the exact argmax—so the measured quantity is the *cost*, not task accuracy.) The reduction was robust across head dimension; the capacity–search tension is a genuine asymptotic trade-off, but it did not bite for $d \geq$ (number of clusters), since query-specific anisotropy needs only $\sim$1 dimension per cluster—real head dimensions (64–256) sit above this, the trade reappearing only as $d \to$ (number of clusters).

Length generalization and staging. See Section 8: 32× the trained length at recall 0.982 for $\sim$800 adaptation steps, SSA-preserved.

Construction pipeline. See Section 9: swap +13.2 perplexity, 94% recovered to within +1.2 of the dense-adapted control at $\sim$38% of keys.

Real-model frozen-swap (Gemma-4-26B). Beyond the 124M control, the swap-and-route was validated end-to-end on a frozen *Gemma-4-26B-A4B* (mixture-of-experts, 4B active): SSA was installed into its five full-attention layers (head dimension 512, $K{=}V$, grouped-query) via the attention interface and scored on needle-in-a-haystack retrieval against the selection budget at $n = 2048$. Plain second-order (cumulant) routing recovers retrieval down to a 50% budget (recall 1.00) but collapses at 25% (recall 0.00); a forward-only probe shows the distant needle's block ranks ≈3rd by the cumulant score—just outside the kept top-2—and worst at the deepest layer, a position-decay effect. Adding the third-cumulant (skew) term of the tempered family (Section 5.2) restores the 50% budget to 1.00 and lifts the 25% budget to 0.44 (to 0.56 if the single worst-routing layer is left dense). This residual is a *tuning artifact*, not a frozen-key ceiling: at a finer block size, plain second-order routing reaches recall 1.00 at the 25% budget—the third-cumulant term is a coarse-block compensator, unnecessary (and at large β harmful) once the needle dominates a small block. So at moderate n the frozen model retrieves fully under the right routing granularity; key co-adaptation (the routability regularizer, (7.2)) is for the long-context regime, where the block-score computation itself becomes quadratic. This is a routing-*quality* measurement (the analytic, score-materializing swap at moderate n), not the subquadratic-kernel regime. The routing survives a doubling of context—a forward-only probe shows the far needle staying within the 25% budget from $n = 2048$ to $n = 4096$, carried by a stable subset of layers. This remains an analytic routing-quality result with no Gemma speed claim. The separate Qwen experiment above supplies complete real-model execution evidence beyond 10M; it does not extend the Gemma result to that length.

Retrieval regime boundaries. The complete 10M run supplies one successful semantic ranking. The controlled sweep below isolates the geometry behind such single-target results by measuring retrieval as a function of context length at fixed budget $\kappa \approx 10^3$ and margin $\Delta = 0.55$:

context n	dense	SSA, isolated target	SSA, benign target
1024	1.00	1.00	1.00
4096	1.00	0.47	1.00
16384	1.00	0.20	1.00
65536	0.90	0.10	1.00
262144	0.83	0.00	1.00

An *isolated* target—a lone spike with no correlated neighbors—*collapses* with length: cheap moment routing averages it into its block and loses it among the fluctuations of the growing number of blocks (the $1/b$ attenuation of Section 5.2, competing against more and more random blocks). This is Proposition 3 in miniature. A *benign* target—one accompanied by a coherent span of query-aligned neighbors, as a real answer is by its surrounding context—lifts its whole block's score and stays flat at 1.00, *beating dense* at long n because selection caps the effective distractor count. The same separation appears across margins: an isolated target barely fires at any margin, while a benign one tracks dense once the margin clears the budget floor $\sqrt{2\log\kappa/d}$. So the headline numbers certify the *easy and benign* regime (single, high-margin, accuracy not losslessness)—the regime everyone agrees is achievable—and do not certify lossless selection in the worst case, which Proposition 3 shows is unavailable cheaply.

12 Discussion and limitations

SSA is best understood as the resolution of a constrained problem rather than a universal accelerator. The recovery-weight law (3.1) shows selection makes retrieval flat in n; the admissible bound (5.1) and the prune gate (5.6) show summaries can certify that selection losslessly; the trilemma (Section 6) shows this can be cheap *only* on benign geometry; and the regularizer (7.2) shows training can supply that geometry. The construction pipeline (Section 9) and the staging ladder (Section 8) turn the mechanism into a recipe that retrofits a dense model and extends its context a rung at a time.

The current limitations follow directly from the theory and evidence. (i) *Worst-case losslessness is not available cheaply* —a sufficiently adversarial or genuinely low-margin target can always evade summary routing; SSA's guarantees are conditional on the (trained, measured) benign geometry. (ii) *Multi-needle and low-margin retrieval* are the hard regime the headline single-needle numbers do not address. (iii) The selection budget κ sets a floor margin $\sqrt{2\log\kappa/d}$; targets below it are missed regardless of n. (iv) The complete 10M experiment uses a frozen 0.5B model, static YaRN far outside its training range, one semantic NIAH ranking, and no dense 10M baseline; it is execution evidence rather than a broad quality result. (v) The 12M IVF near-floor timing is single-head and synthetic, while the real-model 128K timing and quality measurements are smaller-scale. Broad retrieval, perplexity, multi-hop evaluation, trained long-context models, and frontier-model validation remain open. (vi) The complete implementation is not formally verified end to end. Appendix B gives self-contained statements and proofs of the supporting exact-arithmetic invariants; access to the separate formalization is not required to inspect them. As corroborating provenance, private Substrate commits from the potential-store precursor `88a4c7012` through `fad55ff82` machine-check the corresponding results for recursive real-valued ball containment and monotone drop sets, causal prefix-cut selection, conditional 9-vote retention by a 128-slot highest-count reservoir at the 14-selector/70-item route bounds, and survival with a uniform cardinality cap under union with a fixed carrier. It also checks the structural two-pass comparison and its fixed-pass expressivity fence, the strictly-causal finite-transfer identity, and the RoPE winding facts stated below. It also proves that the radial pairing cap is attained under alignment plus a realizable boundary member and that the per-centre refinement agrees there. Those alignment hypotheses are sufficient, not shown necessary. The per-centre cap is universally no larger; a plane witness exhibits a strict gap as large as the whole cap, but no converse says nonalignment forces strictness or equality forces alignment. These results do not formally verify the Python/CUDA mapping, float32 outward rounding, fixed-beam quality, or the unrecorded premise that the measured target had nine pre-consensus base-route votes. The float32 portion is instead addressed experimentally below.

That audit also closes four previously separate proof obligations. First, the restricted-read TV, KL, exact output residual, two block-output bounds, equal-value fence, and smoothed reverse-divergence limit are one checked family. Second, admissible region caps plus strict skip and truncation tests return the strict top set; without strict separation, an explicit index order is needed to name one set. SSA now breaks exact parent-score ties by the larger parent index. Third, the adaptive read-budget ceiling is the grounded-probe theorem stated in Section 6, including its finite-randomized extension and zero-depth counterexample. Fourth, the nine-vote retention, fixed-carrier union, `roundWidth + 128` capacity, past bound, and causal cut are checked as one composed routing plan. The last statement is still conditional on the vote floor and says nothing about attention-output quality, FLOPs, or latency.

The newer formal layer connects those routing facts to a finite score/value store whose softmax read is the derivative of its log-partition and proves the exact append law (5.14). It turns the mean/variance routing statistic into the bounded-range interval (5.4), while explicitly requiring third-cumulant control along the whole tilt segment. It composes state-dependent routed-read errors by the Lipschitz recurrence (5.15), with a counterexample ruling out a final bound from local error alone. It also proves that the pointwise query-group pairing maximum is the least shared safe claim and is invariant under one common isometry, defines the distribution-weighted node-read objective and finite certified-partition choice, and extends the single-spike read ceiling to finite-state indexed and finite-randomized readers. None of these results constructs a cheap summary, learns a partition, supplies the required Lipschitz constants, proves route quality, or turns finite state count into an implementation cost.

The variance-sensitive extension proves (5.16), lifts it through trace spread, arbitrary tree frontiers, causal cuts, and the omitted-mass/output certificate, and gives conditional node/key work counts. Its later full-covariance identity makes $q^T Cq$ the exact directional variance with no dimension factor. The resulting subquadratic statement still assumes a bounded active frontier and the arithmetic work inequality; the Qwen measurement above shows neither trace nor full covariance establishes that premise on the tested tree. The outlier-peel extension proves the order-statistic and recentring radii, exact-exposed-plus-core cap, running-minimum/frontier/causal/output transport, and conditional $O(td)$ work/storage accounting. It does not prove that any fixed peel depth makes the active frontier bounded. Its Inference consumer re-exports these facts without weakening their premises: residual moments and radii are supplied, while subquadraticity still requires bounded active width, the stated certificate charge, and the explicit total-work inequality.

The production tree inflates every recursive norm-plus-child-radius result by $8(d+4)\varepsilon_{32}$ and rounds the candidate and selected maximum toward $+\infty$; query/node score caps receive an analogous absolute-error allowance and upward final rounding. On an RTX 4080, the public verifier compared the actual tree with float64 descendant oracles across five ordinary and adversarial geometries and fanouts 2, 4, and 16. Unguarded arithmetic underestimated 8,701 of 90,105 radii and 367,480 of 1,081,260 score caps. The guarded implementation had zero observed underestimates. At 65,536 leaves, dimension 64, and fanout 16, construction measured 0.433 ms guarded versus 0.223 ms raw. This is empirical stress evidence, not a directed-rounding proof, and it does not turn the fixed-beam traversal into exact top-k search. The complete record is `runs/float_tree_verification.json`; reproduce it with `python -m ssa.float_tree_verification`.

Two adjacent results remain boundaries rather than implementation claims. A positive log-concave score-spread function has a nonincreasing ratio across any fixed nonnegative displacement, but SSA has not established that empirical hypothesis and the result is not an admissible skip bound. Separately, for a strictly below-diagonal linear interaction A on an n-position carrier, $A^n = 0$ and $(I - A)^{-1} = \sum_{k=0}^{n-1} A^k$ without a decay condition. Ordinary causal softmax includes its diagonal and a transformer includes nonlinear stages, so this identity does not make one SSA layer a complete multi-hop solver; it clarifies why the measured two-hop failures remain an empirical model-and-routing question.

The compression comparator has a separate sign boundary. A product of gains in $[0, 1]$ can shrink or erase a stored scalar but cannot reverse its sign; once negative gains are admitted, reversal is determined by their parity. For the DeltaNet-style rank-one correction $T(x) = x - \beta\langle k, x\rangle k$, the key direction has gain $1 - \beta\|k\|^2$ and becomes the exact reflection across $k^\perp$ at $\beta\|k\|^2 = 2$. This explains a capability boundary of the compression arm; it does not improve or certify SSA's selector.

A Derivations

A.1 Recovery weight (3.1). With one target at logit $a_\star$ and μ distractors at $a_\star - \Delta$, $w_\star = e^{\beta a_\star}/(e^{\beta a_\star} + \mu e^{\beta(a_\star - \Delta)})$. Divide numerator and denominator by $e^{\beta a_\star}$ to get $1/(1+\mu e^{-\beta\Delta}) = \sigma(\beta\Delta - \log\mu)$, since $1/(1 + e^{-x}) = \sigma(x)$ with $x = \beta\Delta - \log\mu$.

A.2 Cumulant score (4.2). Let $g(\beta) = \beta^{-1}\log\frac{1}{b}\sum_{j\in c} e^{\beta\langle q,k_j\rangle}$. As $\beta \to 0$, $g(\beta) = \mathbb{E}_j\langle q, k_j\rangle + \frac{\beta}{2}\operatorname{Var}_j\langle q, k_j\rangle + O(\beta^2)$, the cumulant generating function expansion. The mean is $\langle q, \mu_c\rangle$ and the variance is $q^\top\Sigma_c q$, giving (4.2).

A.3 Log-sum-exp sandwich (5.3). For any reals $x_1, \dots, x_b$ with $M = \max_j x_j$: $e^{\beta M} \le \sum_j e^{\beta x_j} \le b\, e^{\beta M}$. Take log, divide by β: $M \le \beta^{-1}\log\sum_j e^{\beta x_j} \le M + \beta^{-1}\log b$.

A.4 Samuelson bound (5.5). Center the data, $d_j = s_j - \bar{s}$, so $\sum_j d_j = 0$. Fix index i. By Cauchy–Schwarz over the other $m - 1$ indices, $d_i^2 = (\sum_{j\ne i} d_j)^2 \le (m-1)\sum_{j\ne i} d_j^2 = (m-1)(\sum_j d_j^2 - d_i^2)$. Hence $d_i^2\, m \le (m-1)\sum_j d_j^2$, i.e. $(s_i - \bar{s})^2 \le (m-1)\operatorname{Var}$. Taking the max over i and adding $\bar{s}$ gives the stated bound on $\max_j s_j$, and (5.6) is its contrapositive against the threshold $s^\star$.

B Self-contained routing invariants

This appendix contains the mathematical content used to justify the 10M router's center-radius hierarchy, bounded top selection, adaptive read-budget boundary, causal selected reads, cross-head reservoir, fixed cross-layer carrier, finite potential and append laws, cumulant intervals, sequential error composition, query-group transport, distribution-weighted safe partitions, finite index capacity, and the variance-sensitive mass tree, plus the factorized-attention and strictly-causal comparisons used to delimit the claims. The restricted-read/output proof is already self-contained in Section 5.7. It is included so the public artifact does not depend on access to the separate Lean repository. The formal audit is useful corroboration, but the definitions, statements, proofs, counterexamples, and scope needed to assess the claims are all below. Every geometric statement is over an exact real inner-product space. The production implementation uses conservative outward inflation and is stress-tested against float64 oracles as reported above; that experiment is not a proof of all floating-point executions.

B.1 Finite families of key regions

Let I be finite and nonempty. In a real inner-product space, let key x_i lie in the closed ball with centre c_i and radius r_i:

$$\|x_i - c_i\| \le r_i.$$

For a reference point p, define the *radial reach* and its score cap by

$$R = \max_{i \in I}(\|c_i - p\| + r_i), \qquad U_R(q) = \langle q, p \rangle + \|q\| R. \tag{B.1}$$

Proposition B.1 (Admissibility and radial minimality)**.** *For every i and q,*

$$\|x_i - p\| \le R, \qquad \langle q, x_i \rangle \le U_R(q).$$

Moreover, R is attained by some index and is the least number satisfying $\|c_i - p\| + r_i \le R$ for all i.

Proof. The triangle inequality gives

$$\|x_i - p\| \le \|x_i - c_i\| + \|c_i - p\| \le r_i + \|c_i - p\| \le R.$$

Then $\langle q, x_i \rangle - \langle q, p \rangle = \langle q, x_i - p \rangle \le \|q\|\|x_i - p\| \le \|q\| R$ by Cauchy–Schwarz. Attainment and minimality follow directly because (B.1) is the maximum of a finite, nonempty family. □

Radial minimality does *not* make U_R the sharpest score bound available from the same data. Define the per-centre cap

$$U_C(q) = \max_{i \in I}(\langle q, c_i \rangle + \|q\| r_i). \tag{B.2}$$

Proposition B.2 (Per-centre refinement)**.** *Every key is bounded by its own centre,*

$$\langle q, x_i \rangle \le \langle q, c_i \rangle + \|q\| r_i,$$

and $U_C(q) \le U_R(q)$ for every query.

Proof. Apply Cauchy–Schwarz to $x_i - c_i$ for the first inequality. For the second, apply it to $c_i - p$:

$$\langle q, c_i \rangle + \|q\| r_i \leq \langle q, p \rangle + \|q\| (\|c_i - p\| + r_i) \leq U_R(q),$$

then maximize the left side over i. □

The comparison can be strict. In $\mathbb{R}^2$, take one region with $p = (0, 0)$, $q = (1, 0)$, $c_1 = (0, 1)$, $r_1 = 0$, and $x_1 = c_1$. Then $R = 1$ and $U_R(q) = 1$, whereas $U_C(q) = \langle q, x_1 \rangle = 0$. Thus the gap can equal the whole reach term. This is one witness, not a theorem that nonalignment always makes the inequality strict.

For the equality case, write $e_q = \|q\|^{-1} q$, using $e_0 = 0$. Say two vectors are on the same ray when one is a nonnegative scalar multiple of the other; then $\langle q, v \rangle = \|q\| \|v\|$.

Proposition B.3 (Aligned realizable attainment)**.** *Suppose an index $i_\star$ attains the reach, $\|c_{i_\star} - p\| + r_{i_\star} = R$, the vectors q and $c_{i_\star} - p$ lie on the same ray, and*

$$x_{i_\star} = c_{i_\star} + r_{i_\star} e_q. \tag{B.3}$$

Then

$$\langle q, x_{i_\star} \rangle = U_R(q).$$

Consequently, if B bounds every $\langle q, x_i \rangle$ for this fixed family and query, then $U_R(q) \leq B$; no smaller skip bound is admissible there. Under the reach-attainment and same-ray hypotheses alone—without the boundary-member hypothesis (B.3)*—$U_C(q) = U_R(q)$.*

Proof. Same-ray equality and (B.3) give

$$\begin{aligned} \langle q, x_{i_\star} \rangle &= \langle q, p \rangle + \langle q, c_{i_\star} - p \rangle + r_{i_\star} \langle q, e_q \rangle \\ &= \langle q, p \rangle + \|q\| (\|c_{i_\star} - p\| + r_{i_\star}) = U_R(q). \end{aligned}$$

Any common bound B must bound this member, proving the next claim. For the cap comparison, the $i_\star$ term in (B.2) equals $U_R(q)$ under alignment, while Proposition B.2 supplies the reverse inequality. □

No $q \neq 0$ hypothesis is needed: at $q = 0$, $e_q = 0$ and both sides vanish. Nor does equality require $r_{i_\star} \geq 0$. That sign condition is needed only to make the constructed boundary member admissible: $\|r_{i_\star} e_q\| \leq r_{i_\star}$ when $r_{i_\star} \geq 0$. The hypotheses above are sufficient; neither attainment nor equality of the two caps is proved to force alignment.

B.2 Recursive ball containment

Define a ball tree recursively. A leaf stores a centre and radius. A node stores a centre c_t and children u, and its radius is

$$\rho_t = \max\left(0, \max_{u \text{ child of } t} (\|c_u - c_t\| + \rho_u)\right). \tag{B.4}$$

Let $s \preceq t$ mean that $s = t$ or s is below t through a finite chain of child edges.

Proposition B.4 (Containment and ancestor cap)**.** *Every child ball lies inside its parent ball. More generally, if $s \preceq t$ and $\|x - c_s\| \le \rho_s$, then*

$$\|x - c_t\| \le \rho_t.$$

For every reference point p, ancestor reach is monotone:

$$\|c_s - p\| + \rho_s \le \|c_t - p\| + \rho_t.$$

Consequently, if each key lies in a descendant ball below t and $\|c_t - p\| + \rho_t \le R_t$, then

$$\langle q, x_i \rangle \le \langle q, p \rangle + \|q\| R_t$$

for every such key.

Proof. Equation (B.4) directly gives $\|c_u - c_t\| + \rho_u \le \rho_t$ for each child. If x lies in u, the triangle inequality yields $\|x - c_t\| \le \|x - c_u\| + \|c_u - c_t\| \le \rho_t$. Induction on the child path proves containment. The same induction, now applying the triangle inequality to $c_u - p$, proves ancestor-reach monotonicity. Proposition B.1 applied at node t gives the score cap. □

For a fixed query and reference point, write $U_v = \langle q, p \rangle + \|q\|(\|c_v - p\| + \rho_v)$. Ancestor-reach monotonicity gives $U_s \le U_t$ whenever $s \preceq t$. Hence, at every threshold θ,

$$U_t \le \theta \quad \Longrightarrow \quad U_s \le \theta.$$

For any finite family of paired ancestor/descendant regions, the descendant drop set therefore contains the ancestor drop set and has at least its cardinality. Likewise Proposition B.2 implies that the per-centre cap drops every region the reach cap drops. These are consequences of bound dominance at a fixed threshold; they do not count keys, traversed nodes, or elapsed work.

This establishes exact-real containment at arbitrary depth. It proves neither that a node radius is minimal nor that the fixed-beam search visits the correct branch. In float32, containment additionally requires radii to be rounded outward or inflated enough to cover accumulated error. The implementation applies the guard measured above at every recursive level and to every score cap; zero observed oracle misses do not promote that guard to a universal arithmetic theorem.

B.3 Uniform and position-dependent orthogonal actions

Proposition B.5 (What orthogonality preserves)**.** *If one orthogonal map A is applied to the query and both keys, then*

$$\langle Aq, Ax\rangle < \langle Aq, Ay\rangle \quad \Longleftrightarrow \quad \langle q, x\rangle < \langle q, y\rangle.$$

Different maps at different key positions need not preserve the order.

Proof. Orthogonality gives $\langle Au, Av\rangle = \langle u, v\rangle$. For the negative statement in $\mathbb{R}^2$, take $q = x = (1, 0)$, $y = (1/2, 0)$, $A = -I$, and $B = I$. Initially $\langle q, y\rangle = 1/2 < 1 = \langle q, x\rangle$, but $\langle q, Ax\rangle = -1 < 1/2 = \langle q, By\rangle$. □

RoPE is position-dependent, so the second statement explains why pre-RoPE and post-RoPE rankings are not identities. It is only an existence counterexample: it says nothing about how often rankings change or which ranking gives better retrieval quality.

B.4 Cross-head consensus and top-count retention

Let H selectors choose finite sets S_h, each of cardinality at most W. Let $U = \bigcup_h S_h$, let $\nu(a) = |\{h : a \in S_h\}|$, and define $C_v = \{a \in U : \nu(a) \geq v\}$.

Proposition B.6 (Consensus counting)**.** *For every v,*

$$|C_v|v \leq HW,$$

and for $v > 0$, $|C_v| \leq \lfloor HW/v \rfloor$.

Proof. Double-count selector–element incidences:

$$\sum_h |S_h| = \sum_{a \in U} \nu(a).$$

The left side is at most HW, while elements of C_v contribute at least $|C_v|v$ to the right side. □

A *top-count reservoir* of capacity k is a set $T \subseteq U$ such that $|T| = \min(k, |U|)$ and every outsider has count no larger than every member:

$$a \in U \setminus T,\ b \in T \quad \Longrightarrow \quad \nu(a) \leq \nu(b). \tag{B.5}$$

Such a reservoir exists by sorting the finite pool by ν; ties may be broken arbitrarily.

Proposition B.7 (Consensus retention)**.** *If $v > 0$ and $\lfloor HW/v \rfloor \leq k$, then every top-count reservoir of capacity k contains C_v.*

Proof. Suppose $a \in C_v \setminus T$. By (B.5), every $b \in T$ has $\nu(b) \geq \nu(a) \geq v$, so $T \cup \{a\} \subseteq C_v$. Hence $|T| + 1 \leq |C_v| \leq \lfloor HW/v \rfloor \leq k$, giving $|T| < k$. The fullness condition $|T| = \min(k, |U|)$ then forces $|T| = |U|$. Since $T \subseteq U$, this implies $T = U$, contradicting $a \notin T$. $\square$

For SSA's route limits, $H = 14$, $W = 70$, and $v = 9$, so

$$|C_9| \leq \left\lfloor \frac{14 \cdot 70}{9} \right\rfloor = 108 < 128.$$

Therefore every nine-vote item is retained by any full 128-slot highest-count reservoir, independent of tie breaking. This is conditional on the item actually receiving nine *pre-reservoir* selections. It says nothing about items at the implementation's two-vote admission threshold, for which the same bound is $\lfloor 980/2 \rfloor = 490$.

The fullness equality in the reservoir definition is essential: merely requiring $|T| \leq k$ allows the empty set, which retains nothing. The multiplied counting bound is attained in general when all H selectors choose the same W items and $v = H$. No construction here is claimed to attain the numeric ceiling 108 at $(H, W, v) = (14, 70, 9)$.

B.5 Accumulating runs and a fixed cross-layer carrier

Let S_ℓ be the selection made at round (or layer) ℓ, and let P be a carried finite set. There are two different constructions:

$$A_0 = P, \qquad A_{n+1} = S_n \cup A_n, \qquad \text{and} \qquad F_\ell = S_\ell \cup P. \tag{B.6}$$

Proposition B.8 (Retention and capacity)**.** *The accumulating run satisfies*

$$A_n = P \cup \bigcup_{\ell < n} S_\ell, \qquad A_n \subseteq A_m \ (n \leq m), \qquad |A_n| \leq |P| + nW$$

when $|S_\ell| \leq W$. If additionally $|P| \leq C$, the fixed-carrier state satisfies, for every ℓ,

$$P \subseteq F_\ell, \qquad |F_\ell| \leq W + C.$$

Proof. The closed form and monotonicity follow by induction from (B.6). The cardinality claims use $|A \cup B| \leq |A| + |B|$, once per induction step for A_n and once directly for F_ℓ. The fixed-carrier inclusion is immediate from the union. $\square$

The constructions must not be conflated. With $P = \varnothing$ and $S_\ell = \{\ell\}$, $F_0 = \{0\}$ is not a subset of $F_1 = \{1\}$, while $A_1 = \{0\} \neq F_1$. Thus the fixed carrier preserves P, not every earlier layer's transient selection, and its $W + C$ cap is uniform precisely because it does not accumulate those selections.

Finally, if "past-bounded at t" means every member of a set is at most t, a union is past-bounded only when both components are. This property is preserved by either construction under the corresponding hypotheses; the union does not create it: at $t = 0$, $P = \{0\}$ is past-bounded but $S_0 = \{5\}$ and $S_0 \cup P$ are not. None of these set identities specifies how a selector admits an item, proves a causal mask, or proves that the measured target belonged to P.

Combining consensus retention, the fixed-carrier bound, and the position cut gives the concrete composed plan used by SSA. If 14 heads each select at most 70 blocks, P is a full highest-count reservoir of capacity 128, every later selection S_ℓ has size at most `roundWidth`, and both P and S_ℓ lie at or before the query position, then every nine-vote block lies in P and in $S_\ell \cup P$; that union has size at most `roundWidth + 128`, remains past-bounded, and its position-cut read is causal. This conjunction is conditional on the nine-vote premise. A 140-block pooled family in which every block has at most two votes shows that a full 128-slot reservoir can drop a below-threshold block; capacity alone does not supply the premise.

B.6 Selected reads and position causality

Let L be linearly ordered and let $S \subseteq L$ be a finite routed set. Define the whole-set and position-cut reads

$$W_S(i) = S, \qquad C_S(i) = \{j \in S : j \le i\}. \tag{B.7}$$

Call a read R causal when $j \in R(i)$ always implies $j \le i$.

Proposition B.9 (Causal cut and composition)**.** *The cut read C_S is causal for every S. The whole-set read W_S is non-causal whenever some selected j lies after a query i; in particular, any S with two distinct elements fails at its least element. A composition of causal relations is causal.*

Proof. Membership in $C_S(i)$ includes $j \le i$ by definition. For W_S, the later selected position is returned at the earlier query. Finally, if a first causal stage relates i only to $k \le i$ and a second relates k only to $j \le k$, transitivity gives $j \le i$. □

Chunk causality is not position causality: positions 0 and 1 in the same chunk have equal chunk index, yet a whole-chunk read at position 0 may expose position 1. SSA's token-level mask implements C_S, including at a partially visible boundary block.

B.7 Two-pass cover and its expressivity fence

Index $n = pw$ positions by pairs (a, u) of a block a and slot u. Let a local relation connect pairs with the same block, and a global relation connect pairs with the same slot.

Proposition B.10 (Complete two-hop cover and balanced scalar cost)**.** *Every source (a, u) reaches every target (b, v) by local then global steps through (a, v); the reverse order works through*

(b, u). *Neither relation alone is complete when* $p, w \geq 2$. *If an external accounting assigns cost proportional to* $n(w + n/w)$ *with real* $w > 0$, *then*

$$w + \frac{n}{w} \geq 2\sqrt{n}, \tag{B.8}$$

with equality exactly at $w = \sqrt{n}$.

Proof. The displayed intermediates share the required coordinate with each endpoint. Pairs differing in both coordinates witness the failure of either relation alone. For the cost, $(w - \sqrt{n})^2/w = w + n/w - 2\sqrt{n} \geq 0$; equality of a square holds exactly at the stated point. □

Connectivity is not functional equivalence. Give each block an arbitrary $w \times w$ local mixing matrix and each slot an arbitrary $p \times p$ global mixing matrix. With the local family fixed, the composite depends linearly on wp^2 global parameters, whereas all linear maps on the pw tokens form a space of dimension p^2w^2. Thus for $p \geq 1, w \geq 2$ the reachable family is a strict subspace; symmetrically, fixing the global family gives at most $pw^2 < p^2w^2$ dimensions when $p \geq 2, w \geq 1$. This says nothing about the bilinear family when both passes vary and implies no rank bound: two differently blocked full-rank factors can have a full-rank product.

B.8 Strictly causal finite transfer

Let A be an $n \times n$ matrix with $A_{ij} = 0$ unless $j < i$—a strictly past-only linear interaction with no diagonal.

Proposition B.11 (Nilpotence and exact finite inverse)**.** *A nonzero entry of* A^k *can move at least* k *places down the order. Consequently* $A^n = 0$ *and*

$$(I - A)^{-1} = I + A + \cdots + A^{n-1}, \tag{B.9}$$

with no bound on the magnitudes of A*'s entries.*

Proof. Induct on k. Each additional matrix multiplication inserts one strictly increasing intermediate index, so a length-k path needs at least k distinct order steps. No such path of length n exists on n positions, hence $A^n = 0$. Multiplying the finite geometric sum by $I - A$ on either side telescopes to $I - A^n = I$. □

This applies to a linear strictly-below-diagonal operator. Standard causal attention admits the current position, and transformer layers contain normalization and nonlinear maps, so (B.9) is not a one-layer multi-hop guarantee for SSA.

B.9 Rotary winding and extrapolation

Let a lifted phase schedule $\theta_0, \ldots, \theta_N \in \mathbb{R}$ close after k turns, $\theta_N - \theta_0 = 2\pi k$, and suppose every step has magnitude at most ω.

Proposition B.12 (Wavelength floor and winding stability). *The turn count obeys*

$$2\pi|k| \leq N\omega. \tag{B.10}$$

Thus $N\omega < 2\pi$ forces $k = 0$. If two closed lifted schedules θ, ϕ differ by less than π at every sampled position, then their turn counts agree.

Proof. Telescope the increments and apply the triangle inequality: $2\pi|k| = |\sum_{t<N}(\theta_{t+1} - \theta_t)| \leq N\omega$. For stability, the difference of the two endpoint differences is $2\pi(k - \ell)$, while it is also $(\theta_N - \phi_N) - (\theta_0 - \phi_0)$, whose magnitude is strictly below 2π; the only such integer multiple of 2π is zero. □

For angles supplied only modulo 2π, identifying the sampled discrete winding with a lift additionally requires an anti-aliasing choice—no step may cross half a turn. These statements delimit a positional regime; they do not prove that attention quality is good within it or bad outside it.

B.10 Two analytic baselines not used as certificates

Proposition B.13 (Log-concave spread ratio). *If $\sigma : \mathbb{R} \to (0, \infty)$ has concave $\log \sigma$, then for every $c \geq 0$ the ratio $\sigma(x + c)/\sigma(x)$ is nonincreasing in x.*

Proof. A concave function has nonincreasing increments across a fixed nonnegative displacement, so $\log \sigma(x + c) - \log \sigma(x) \geq \log \sigma(y + c) - \log \sigma(y)$ for $x \leq y$. Exponentiation preserves the order. □

This can support a representative-position heuristic only after the score-spread shape is measured. It is not an upper bound on any individual key and licenses no exact prune. The converse at one displacement is false, and a conclusion measured at one block length need not transfer to another.

Proposition B.14 (Ideal scalar tree fanout). *For $B > 1$, in the model that charges f units at each of $\log_f B$ levels, $f \log_f B$ is minimized for real $f > 1$ at e and for integer $f \geq 2$ at 3.*

Proof. Apart from the positive constant $\log B$, differentiate $f/\log f$; its derivative has the sign of $\log f - 1$, so the unique real minimum is e. The function increases for integers $f \geq 3$, and $3/\log 3 < 2/\log 2$ is equivalent to $2^3 < 3^2$. □

This scalar model omits the costs and quality effects that dominate a batched GPU tree. It motivates a fanout sweep; it does not select SSA's production fanout.

B.11 Gain signs and the rank-one reflection corner

Let $G_m = \prod_{t<m} a_t$. If every $a_t \in [0, 1]$, induction gives $G_m \in [0, 1]$, so multiplying a positive stored scalar by G_m cannot make it negative. If every factor is nonzero, the sign of G_m is negative exactly when an odd number of factors are negative.

For a nonzero k define the rank-one correction

$$T_{\beta,k}(x) = x - \beta\langle k, x\rangle k. \tag{B.11}$$

It fixes every $x \perp k$ and sends $k \mapsto (1 - \beta\|k\|^2)k$. Therefore the key line reverses exactly when $\beta\|k\|^2 > 1$, and at $\beta\|k\|^2 = 2$ the map is $+1$ on $k^\perp$ and -1 on the line spanned by k: precisely the orthogonal reflection across $k^\perp$. These are statements about a compressed linear state update, not sparse selection or attention quality.

B.12 Bounded selection and the meaning of "top"

Let a finite carrier I be partitioned into regions by $b : I \to C$, let s_i be its scores, and suppose $s_i \le U_{b(i)}$ for admissible region caps U_c. A search probes regions P, returns $R \subseteq I$, and uses a threshold τ.

Proposition B.15 (Strict bounded selection)**.** *Assume every unprobed cap satisfies $U_c < \tau$, every unreturned member of a probed region has score below τ, and every returned member has score at least τ. Then*

$$R = \{i : s_i \ge \tau\},$$

and every outsider scores strictly below every member of R. If additionally $|R| = k$, then R is a strict top-k set.

Proof. A returned member is in the displayed set by hypothesis. An unreturned member is either in a probed region and covered by the direct truncation condition, or in an unprobed region and has $s_i \le U_{b(i)} < \tau$. This proves the reverse inclusion. Combining an outsider's strict upper bound with a member's lower bound gives strict separation. □

The count $|R| = k$ is a hypothesis, not a consequence of admissibility. Nor does a weak score order name a unique set at a tie: under two equal scores, either singleton is weakly top one and neither is strictly top one. A deterministic repair orders equal scores by index. SSA uses the larger parent index as the winner of an exact routing-score tie. The indexed CCC certificate uses strict unprobed-cap and search-truncation tests; its exhaustive corner uses this explicit tie policy.

B.13 Adaptive read budgets

Represent a deterministic selector as a finite binary decision tree. Each internal node probes one of n positions; on the input with its unique spike at j, the answer is whether that node probes j.

Each leaf returns a set of positions. The read depth is the maximum root-to-leaf probe count, and the selector is *grounded* when every returned position belongs to that input's probe trace.

Proposition B.16 (Grounded recall ceiling)**.** *A grounded selector of read depth at most b recalls at most b of the n spike placements. Under a uniform placement its recall is at most b/n. For every finite distribution over grounded selectors of depth at most b, some fixed placement has seed-averaged recall at most b/n.*

Proof. On the all-false reference path the trace E has at most b members. If $j \notin E$, planting the spike at j changes no answer on that path, so the trace and returned set remain the reference ones. Grounding then prevents the returned set from containing j. Thus the recalled placements form a subset of E. For a finite randomized family, sum the reach indicator over placements and seeds in either order. Every seed's sum is at most b, so the average total is at most b and some placement has average at most b/n. $\square$

Grounding is necessary: the depth-zero leaf that returns all n positions recalls every placement. A reader that probes any fixed set E and returns exactly E attains the bound, so the count is sharp. Preprocessed indexes with side information outside the probe trace are not modeled by this proposition.

B.14 Potential reads and append-only stores

Let I be a nonempty finite set, let $a_i(q) \in \mathbb{R}$ be probe-dependent scores, and let v_i be stored values in a real normed vector space. Define

$$Z(q) = \sum_{i \in I} e^{a_i(q)}, \quad \Phi(q) = \log Z(q), \quad p_i(q) = \frac{e^{a_i(q)}}{Z(q)}, \quad r(q) = \sum_i p_i(q) v_i.$$

These are definitions from the score/value store, not four independently supplied operations.

Proposition B.17 (One-potential read and exact append law)**.** *For scalar v_i,*

$$\frac{d}{dt} \log \sum_i e^{a_i(q)+tv_i} \bigg|_{t=0} = r(q).$$

If a nonempty finite store of mass Z and read r receives one fresh site x of mass $a = e^{a_x} > 0$ and value v_x, then

$$Z' = Z + a, \qquad \Phi' - \Phi = \log(1 + a/Z), \qquad p'_i = \frac{Z}{Z+a} p_i \ (i \neq x), \qquad r' - r = \frac{a}{Z+a}(v_x - r).$$

Proof. Differentiate the finite exponential sum and divide by it to obtain $\sum_i e^{a_i} v_i / Z$. The mass identity is finite-sum insertion. Factor $Z + a = Z(1 + a/Z)$ for the log identity and divide each old numerator by the new denominator for the weight identity. Finally, $r' = (Zr + av_x)/(Z + a)$, whose difference from r is the displayed update. $\square$

When the score surface changes at the same round, insert and subtract the read (or log mass) of the old carrier evaluated with the new scores. This splits total change exactly into the append term at the new scores and a score-only term on the unchanged old carrier. Consequently append-only retention does not imply access-weight retention: old weights strictly fall at a pure append, but no output-degradation direction, cumulative drift bound, recovery rule, or convergence statement follows.

B.15 Bounded-interval cumulant routing

For logits $x_i \in [\ell, h]$, define $K(b) = \log(|I|^{-1} \sum_i e^{bx_i})$, their uniform mean μ, and their uniform population variance σ^2. Let $\mathbb{E}_b$ denote expectation under weights proportional to e^{bx_i}.

Proposition B.18 (Second-order routing interval)**.** *For every real b,*

$$\left| K(b) - \left(b\mu + \frac{b^2\sigma^2}{2} \right) \right| \le \frac{|b|^3 (h - \ell)^3}{6}.$$

More generally, $(h - \ell)^3$ may be replaced by any M satisfying $|\mathbb{E}_c[(x - \mathbb{E}_c x)^3]| \le M$ for every c between 0 and b.

Proof. Direct differentiation of the finite log-partition gives $K'(c) = \mathbb{E}_c x$, $K''(c) = \mathbb{E}_c[(x - \mathbb{E}_c x)^2]$, and $K'''(c) = \mathbb{E}_c[(x - \mathbb{E}_c x)^3]$. At zero these are μ and σ^2. Every tilted mean lies in $[\ell, h]$, hence $|x_i - \mathbb{E}_c x| \le h - \ell$ and $|K'''(c)| \le \mathbb{E}_c|x - \mathbb{E}_c x|^3 \le (h - \ell)^3$. Taylor's theorem with Lagrange remainder gives the claim. $\square$

This encloses a supplied block's normalized log-sum-exp; it neither selects a route nor proves retrieval accuracy. The ordinary third cumulant at zero alone does not satisfy the segment hypothesis.

B.16 Sequential error composition

Let $F_t, G_t : E \to E$ be exact and approximate maps on a metric space. Assume F_t is L_t-Lipschitz and $d(G_t(x), F_t(x)) \le \epsilon_t$ for every input. Starting both paths at x_0, write $e_t = d(\tilde{x}_t, x_t)$.

Proposition B.19 (Lipschitz-weighted multi-hop error)**.** *The paths obey*

$$e_{t+1} \le \epsilon_t + L_t e_t,$$

and hence the final error is bounded by the fold beginning at zero and repeatedly applying $z \mapsto \epsilon_t + L_t z$. If every $L_t \le 1$, then $e_N \le \sum_{t<N} \epsilon_t$; if $F_t = G_t$ everywhere, the paths agree exactly.

Proof. Insert $F_t(\widetilde{x}_t)$ between $G_t(\widetilde{x}_t)$ and $F_t(x_t)$. The triangle inequality bounds the first distance by ϵ_t, and Lipschitzness bounds the second by $L_t e_t$. Induction gives the fold and the nonexpansive sum. Pointwise equality gives exactness by the same induction. $\square$

Gain control is necessary. Given any proposed two-step bound B, let the first approximate scalar map differ from the exact one by one and let the shared second map multiply by $|B|+1$. The second local error is zero, yet the final discrepancy is $|B|+1 > B$. Thus local read certificates alone do not certify a multi-hop path.

B.17 Query groups and uniform route transport

For finite nonempty query and part sets Q, B with vectors q_s, x_c, define the shared claim $U^*(c) = \max_{s\in Q}\langle q_s, x_c\rangle$. Call U safe when $\langle q_s, x_c\rangle \le U(c)$ for every s, c.

Proposition B.20 (Least shared claim)**.** *A function U is safe exactly when $U^*(c) \le U(c)$ for every c; thus U^* is the least shared safe bound. If $\theta < \max_c\langle q_s, x_c\rangle$ for every s, then $\{c : U^*(c) > \theta\}$ contains a maximizing part for every query. Adding another query can only shrink the drop set $\{c : U^*(c) \le \theta\}$. If one linear isometry A is applied to every q_s and x_c, then U^*, its retained set, and the safety relation are unchanged.*

Proof. A finite maximum dominates every member and is below every common upper bound, proving safety and minimality. A query maximizer has score above θ and no larger than U^*, so it is retained. Enlarging the query family can only raise a pointwise maximum. Finally, $\langle Aq_s, Ax_c\rangle = \langle q_s, x_c\rangle$ for a common isometry, so every displayed object is fixed. $\square$

This supplies no cheaper query summary: evaluating the exact top claim may itself cost $|Q|$ pairings per part. Nor does it cover distinct position-dependent maps, which can reverse pairing order.

B.18 Distribution-sensitive certified partitions

For a finite query set with nonnegative weights w_q, let the unit-read cost of a count profile r_q be $C(r) = \sum_q w_q r_q$. A safe partition assigns every item i to a cell $c(i)$ and supplies bounds $U(q, c)$ with $s(q, i) \le U(q, c(i))$. At threshold $\tau(q)$ it reads cells with $U(q, c) > \tau(q)$.

Proposition B.21 (Safe finite-family choice)**.** *Pointwise $r_q \le s_q$ implies $C(r) \le C(s)$; if the inequality is strict at a query of positive weight, so is the cost inequality. A cell not read by a safe partition contains no item above threshold. Replacing its bounds by pointwise lower safe bounds reads a subset of cells and cannot increase weighted cost. Every supplied finite nonempty family of safe partitions has a least-cost member. If a tree traversal and a partition read equally many nodes or cells for every query, their weighted costs agree.*

Proof. Each cost summand is monotone because $w_q \geq 0$, and a strict positive-weight summand makes the finite sum strict. For an omitted cell, $U(q,c) \leq \tau(q)$ and safety gives $s(q,i) \leq \tau(q)$. Lower bounds can only remove members from $\{c : U(q,c) > \tau(q)\}$; cardinality and the first monotonicity result give the cost claim. A real-valued function on a finite nonempty candidate family attains a minimum. The final statement substitutes equal pointwise counts into the same finite sum. □

The result chooses among supplied certified candidates. It constructs no partition or tree, proves no global optimum or generalization, and treats one node read as one cost unit rather than wall-clock work.

B.19 Finite index capacity

Extend the spike model above with a finite index-state set C, $|C| = K$. Each spike placement j selects a state $c(j)$, and that state selects an adaptive read tree. Suppose every state tree has depth at most b and its all-false reference output has at most a members.

Proposition B.22 (Indexed and randomized ceiling)**.** *The indexed reader recalls at most $K(b+a)$ spike placements. If every state tree is grounded, it recalls at most Kb. For any finite distribution over such indexed plans, some fixed placement has seed-averaged recall at most $K(b+a)/n$ and miss share at least $1 - K(b+a)/n$. Under grounding the same averaging argument tightens the recall ceiling to Kb/n.*

Proof. For one fixed state, the reference-path argument of Proposition B.16 shows that every successful placement lies either in its reference probe trace or in its reference output, a set of size at most $b + a$. The complete indexed success set is contained in the union of these sets over K states, so its cardinality is at most $K(b + a)$. Grounding puts the reference output inside the trace and removes a. For randomization, sum reach indicators first over positions and then seeds; every seed contributes at most $K(b + a)$. Reversing the two finite sums and averaging forces some position below the mean, and reach plus miss equals one. □

The dependence on K cannot be removed: the identity index takes $C = \{0, \ldots, n-1\}$, stores j, and selects a depth-zero leaf returning $\{j\}$. It recalls every spike. State count is not a storage-bit or runtime cost, and the theorem does not cover scored, approximate, or multiple-target retrieval.

References

[1] A. Vaswani et al. *Attention Is All You Need.* NeurIPS, 2017.

[2] R. Child, S. Gray, A. Radford, I. Sutskever. *Generating Long Sequences with Sparse Transformers.* 2019.

[3] I. Beltagy, M. E. Peters, A. Cohan. *Longformer: The Long-Document Transformer.* 2020.

[4] M. Zaheer et al. *Big Bird: Transformers for Longer Sequences.* NeurIPS, 2020.

[5] N. Kitaev, Ł. Kaiser, A. Levskaya. *Reformer: The Efficient Transformer.* ICLR, 2020.

[6] A. Roy, M. Saffar, A. Vaswani, D. Grangier. *Efficient Content-Based Sparse Attention with Routing Transformers.* TACL, 2021.

[7] A. Katharopoulos, A. Vyas, N. Pappas, F. Fleuret. *Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention.* ICML, 2020.

[8] K. Choromanski et al. *Rethinking Attention with Performers.* ICLR, 2021.

[9] J. Su et al. *RoFormer: Enhanced Transformer with Rotary Position Embedding.* 2021.

[10] L. Greengard, V. Rokhlin. *A Fast Algorithm for Particle Simulations.* J. Comput. Phys., 1987.

[11] J. Barnes, P. Hut. *A Hierarchical $O(N \log N)$ Force-Calculation Algorithm.* Nature, 1986.

[12] P. A. Samuelson. *How Deviant Can You Be?* J. Amer. Statist. Assoc., 1968.

[13] A. Keles, P. Wijewardena, C. Hegde. *On the Computational Complexity of Self-Attention.* ALT, 2023.

[14] J. Alman, Z. Song. *Fast Attention Requires Bounded Entries.* NeurIPS, 2023.

[15] H. Ramsauer et al. *Hopfield Networks Is All You Need.* ICLR, 2021.

[16] S. Arora et al. *Zoology: Measuring and Improving Recall in Efficient Language Models.* 2023.

[17] J. Smirl. *Substrate Abstract Algebra*, private Lean 4 development. Source theorems are machine-checked and axiom-pure; the SSA instantiations and numerical implementation are not formally verified. The public mathematical statements and proofs used for the 10M routing invariants are reproduced in Appendix B, so access to this development is not required. Audit mappings and declaration names are recorded in `docs/substrate_math_imports.md`.